A REAL-TIME RESEARCH PROJECT REPORT ON

ANALYZING E- COMMERCE SHIPMENT DELIVERY AND CUSTOMER SATISFACTION

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in the partial fulfilment for the award of BACHELOR OF TECHNOLOGY in INFORMATION TECHNOLOGY during the academic year.

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ANALYZING E- COMMERCE SHIPMENT DELIVERY AND CUSTOMER SATISFACTION

ABSTRACT

E-commerce continues to surge globally, with Statista reporting over 5.3 trillion USD in global e-commerce sales in 2022, projected to reach 8.1 trillion USD by 2026. However, despite rapid growth, around 65% of customers abandon online purchases due to unsatisfactory delivery services or delays.

The current manual and semi-automated shipment tracking systems suffer from inconsistent data recording, lack of real-time customer feedback integration, and limited insight into delivery performance drivers, which leads to poor customer satisfaction.

To address these challenges, this study proposes an Exploratory Data Analysis (EDA) framework focusing on shipment data and customer reviews to identify trends, anomalies, and key delivery satisfaction factors. Using visualizations such as histograms of delivery delays, box plots for shipment time by region, and heatmaps of satisfaction vs. shipment attributes, the analysis unveils meaningful patterns in the logistics pipeline. Further, feature extraction and dimensionality reduction using Fast Independent Component Analysis (FastICA) is conducted to isolate hidden independent factors affecting customer satisfaction, such as courier performance, packaging quality, and order accuracy. Scatter plots and 3D component visualizations of the ICA results offer deeper insight into the underlying features that most significantly influence user ratings, thus paving the way for intelligent logistics optimization and enhanced customer satisfaction model.

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ABBREVATIONS

ICA Independent Component Analysis

EDA Exploratory Data Analysis

IQR Interquartile Range

KDE Kernal Density Estimation

ECDF Empirical Cumulative Distribution Function

CSAT Customer Satisfaction Score

UML Unified Modeling Language

DFD Data Flow Diagram

ESCAP Economic and Social Commission for Asia and the Pacific

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

The e-commerce sector has witnessed exponential growth in recent years. According to Statista, global e-commerce sales reached over $5.3 trillion in 2022 and are projected to surpass $8.1 trillion by 2026. This surge is driven by increased internet penetration, smartphone usage, and consumer preference for online shopping [1][2].

Fig. 1.1: E-commerce order fulfilment process.

Despite this growth, customer satisfaction remains a critical concern [3]. A survey by Celigo revealed that 88% of U.S. and 79% of U.K [4][5][6]. online shoppers experienced disappointments with online retailers in the past year, citing issues like late deliveries and high shipping costs. Furthermore, 48% of consumers reported one to three disappointing shopping experiences annually, with 25% experiencing four to nine such instances [7].

These statistics underscore the importance of efficient logistics and customer service in e-commerce [8][9]. Companies must address these challenges to retain customers and maintain a competitive edge [10][11]. Implementing advanced data analytics can provide insight into

customer behaviour and logistics performance, enabling businesses to make informed decisions and improve service quality [12].

1.2Research Motivation

Real-time companies like Amazon and Alibaba have set high standards in e-commerce logistics by leveraging data analytics to optimize their operations. Amazon, for instance, uses predictive analytics to forecast demand, manage inventory, and streamline delivery routes, resulting in faster delivery times and reduced costs. Similarly, Alibaba employs intelligent supply chain management to enhance supplier performance and reduce inventory costs.

The need for data analysis in these companies stems from the complexity of managing vast amounts of data generated from customer interactions, transactions, and logistics operations. By analyzing this data, businesses can identify patterns, predict trends, and make data-driven decisions to improve efficiency and customer satisfaction. For example, analyzing delivery performance metrics can help identify areas for improvement, leading to enhanced reliability and customer trust.

Moreover, data analytics enables companies to personalize customer experiences by understanding preferences and behaviors. Personalized recommendations and targeted marketing campaigns can increase customer engagement and loyalty. In a competitive market, leveraging data analytics is essential for e-commerce companies to scale effectively and meet evolving customer expectations.

1.3 Problem Statement

The exponential growth of the e-commerce sector, with global sales exceeding $5.3 trillion in 2022 and projected to reach $8.1 trillion by 2026, underscores its pivotal role in the modern economy. Despite this rapid expansion, a significant challenge persists: widespread customer dissatisfaction stemming from inadequate delivery services. Reports indicate that approximately 65% of online customers abandon purchases due to unsatisfactory delivery or delays, and a high percentage of shoppers frequently experience disappointments related to late deliveries and excessive shipping costs.

The root of this problem lies in the inherent limitations of current e-commerce shipment tracking and logistics systems. Many businesses rely on manual or semi-automated processes characterized by inconsistent data recording, a notable absence of real-time customer feedback integration, and insufficient analytical capabilities. This deficiency hinders the identification of key performance drivers and bottlenecks within the logistics pipeline. Consequently, companies lack actionable insights into factors most significantly influencing customer satisfaction, such as courier performance, packaging quality, and order accuracy. Without a robust, data-driven approach to understand these complex interdependencies, e-commerce businesses struggle to proactively address delivery issues, optimize their supply chains, and foster long-term customer loyalty in an increasingly competitive market. This research aims to address this critical gap by providing a systematic analytical framework to transform raw shipment and customer feedback data into actionable intelligence.

1.4 Objective

The overarching objective of analyzing e-commerce shipment delivery customer satisfaction is to understand, measure, and improve the delivery experience to enhance overall customer satisfaction, foster loyalty, and drive business success for online retailers.

- **Identify Key Drivers of Satisfaction**: To pinpoint which specific aspects of the shipment delivery process (e.g., speed, reliability, tracking accuracy, communication, packaging, delivery personnel professionalism, cost of delivery) most significantly impact customer satisfaction.
- **Evaluate Current Performance:** To assess how well existing delivery services are meeting customer expectations and to identify areas of strength and weakness in the current delivery ecosystem.
- **Understand Customer Expectations and Preferences**: To gain insights into what customers truly value in terms of delivery, considering different demographics, product types, and purchasing occasions. This includes understanding their willingness to pay for certain delivery options or wait for longer periods for lower costs..
- **Inform Strategic Decision-Making**: To provide data-driven insights that can guide e-commerce businesses and logistics providers in optimizing their delivery strategies, investing in new technologies, improving operational efficiency, and designing customer-centric delivery options.

- **Optimize Logistics and Supply Chain Operations**: To leverage customer feedback to refine internal logistics processes, improve inventory management, optimize routing, and enhance the overall efficiency of the supply chain.

By achieving these objectives, businesses can transform their delivery service from a mere operational necessity into a significant competitive differentiator and a powerful tool for customer retention and growth.

1.5 Advantages

Here are the needs and significance of your project "E-commerce Shipment and Customer Satisfaction," presented in a format similar to your example:

- **Addressing Critical Customer Churn:** The project is needed to tackle the significant issue of customer abandonment due to unsatisfactory delivery experiences, a problem currently exacerbated by inconsistent data and a lack of real-time insights. This is significant for retaining customers and safeguarding revenue in the competitive e-commerce landscape.
- **Unlocking Actionable Logistics Insights:** It addresses the urgent need for a deeper understanding of the factors driving delivery satisfaction beyond superficial metrics. By employing advanced analytical techniques like EDA and FastICA, the project is significant for uncovering hidden patterns related to courier performance, packaging quality, and order accuracy, enabling targeted operational improvements.
- **Enabling Data-Driven Logistics Optimization:** The proposed framework is crucial for moving beyond reactive problem-solving in e-commerce logistics. By providing clear, data-backed insights into delivery performance drivers, it is significant for optimizing resource allocation, streamlining supply chains, and reducing operational costs, fostering greater efficiency.
- **Elevating Overall Customer Experience and Loyalty:** The project fills the gap in understanding how shipment processes directly impact customer sentiment. By isolating key satisfaction factors, it is significant for allowing businesses to tailor delivery strategies, enhance transparency, and deliver superior, personalized experiences, thereby boosting customer loyalty and positive brand perception.
- **Empowering Proactive Business Management:** By providing a comprehensive analytical framework, the project addresses the need for better tools for e-commerce

businesses. This is significant for empowering stakeholders to proactively manage delivery expectations, forecast demand with greater accuracy, and adapt swiftly to market changes and customer feedback, moving from reactive to predictive operations.

- **Fostering Sustainable E-commerce Growth:** This research is needed to provide a pathway for e-commerce companies to grow sustainably by minimizing shipping-related dissatisfaction. It is significant for reducing negative reviews and returns, improving brand reputation, and ensuring long-term viability by aligning operational efficiency with paramount customer expectations.

1.6Applications

- **Inventory Management**: Predictive analytics helps forecast demand accurately, ensuring optimal inventory levels and reducing the risk of stockouts or overstocking.
- **Supply Chain Optimization**: Analyzing data from various points in the supply chain enables businesses to identify inefficiencies and implement improvements for smoother operations.
- **Customer Segmentation**: Data analysis allows for the segmentation of customers based on behaviour and preferences, facilitating personalized marketing and product recommendations.
- **Fraud Detection**: Monitoring transaction data helps in identifying suspicious activities, enabling timely intervention to prevent fraud and enhance security.
- **Performance Monitoring**: Continuous analysis of key performance indicators (KPIs) assists in tracking progress, setting benchmarks, and making informed strategic decisions.

CHAPTER 2

LITERATURE SURVEY

1.1 Introduction

Logistics is one of the most important means to improve the efficiency of the flow of materials, thereby reducing the distribution costs of the business. In addition to price, quality is also a factor creating competitive advantages which makes quality management one of the most important steps to achieve profit goals for commercial enterprises

2.2 Related Work

[1]. **Fisk, R. et al** For quality management in general, it is impossible to ignore logistics service quality management. The research team led by Stank has shown a positive relationship between logistics service quality and customer satisfaction in the fast-food industry examined the weaknesses of existing e-commerce shipment delivery systems, which traditionally focused on transactional efficiency and operational cost-cutting, often at the expense of the overall customer experience. proposed a service recovery-focused delivery model that integrates proactive problem resolution, customer-centred communication, and structured service recovery protocols into the logistics process .The implementation of such a system led to improved customer trust, higher satisfaction levels, and stronger customer loyalty, even when delivery issues occasionally occurred, because customers felt their concerns were addressed promptly and professionally.

[2]. **Stank, T.P et al** The trend of converting consumers' behaviour from direct to online shopping is increasingly popular, most clearly shown during the COVID-19 epidemic in 2020. In Vietnam, this market is estimated at over USD 20 billion in 2023 with an average growth rate of 29% in the period 2020–2025 according to forecasts of Google (Mountain View, CA, USA), Temasek (Singapore) and Bain & Company (Boston, MA, USA). Specifically, the Generation Z customer group is considered to have the highest online purchasing power because by 2025, Generation Z will account for about 25% of the national workforce, and equivalent to about 15 million VND potential consumers. Specifically, in Hanoi—the capital of Vietnam—88% of Generation Z people have online shopping activities

[3]. **Ha, N.T. et al** Therefore, optimizing e-logistics activities to meet the needs of this customer group is an important mission to complete Many studies have explored the positive influence of e-LSQ on customer satisfaction in the purchase process and customer loyalty

examined the persistent problems in existing e-commerce shipment delivery systems, particularly the lack of coordination between order management, inventory visibility, and delivery execution. proposed an integrated order fulfilment and delivery synchronization model that connects inventory systems, order processing, and logistics providers on a unified real-time platform. result, the implementation of this integrated model significantly reduced order processing times, improved inventory accuracy,

[4,]. **Rao, S. et al** Moreover, many other studies also show a positive relationship between customer satisfaction and loyalty. analyzed the inefficiencies in traditional e-commerce shipment delivery systems, which often prioritized cost minimization and process automation while neglecting the personalization of delivery experiences. proposed a personalized logistics service model that integrates customer preferences, predictive analytics, and flexible delivery arrangements. The implementation of this system led to enhanced customer engagement, improved delivery accuracy, reduced service failures, and significantly higher customer satisfaction

[5] **Griffis, S.E..**et al (2012) investigated the impact of delivery performance on customer satisfaction in online retail. Specifically, they examined how factors like on-time delivery, analyzed the inefficiencies present in existing e-commerce shipment delivery systems, where the focus was heavily on operational performance indicators such as delivery speed and cost, while often overlooking the psychological and behavioural aspects of customer satisfaction. proposed a customer experience-driven logistics model that integrates both operational efficiency and customer engagement throughout the delivery process. The results showed that incorporating these customer-centric elements led to improved customer perceptions of reliability, trustworthiness, and overall satisfaction

[6,]. **Chen, S.-C** et al Journal of retailing and consumer services, 2012 - Elsevier
of customer satisfaction and loyalty by providing insights into the factors that influence investigated the limitations of existing e-commerce shipment delivery systems, which often failed to address the full customer experience by focusing primarily on delivery speed while neglecting service quality, customer interaction, and post-delivery support. proposed an integrated service quality model that combined efficient logistics operations with enhanced customer service elements. The results showed that companies adopting this model experienced improvements not only in operational efficiency but also in customer satisfaction, as customers valued the transparency, responsiveness, and personalized attention they received throughout the delivery process.

[7] **Restuputri, D.P. et al** I focuses on the impact of logistic service quality on customer satisfaction and loyalty, particularly during the COVID-19 pandemic, using Kansei Engineering. examined the challenges in existing e-commerce shipment delivery systems, which frequently suffered from poor last-mile delivery coordination, inefficient routing, and inadequate utilization of delivery resources. proposed an optimized last-mile delivery model that utilized route optimization algorithms, real-time traffic data, and dynamic vehicle scheduling to improve delivery efficiency. The results demonstrated that implementing such optimization strategies led to reduced delivery times,

[8] **Bayles, D.L. et al** shows that the basic factors leading to purchase satisfaction do not have a direct impact on customer retention. analyzed the weaknesses in existing e-commerce shipment delivery systems, where many companies prioritized cost reduction and operational efficiency over customer service quality. proposed a customer-focused delivery framework that integrates proactive communication, customer choice in delivery scheduling, and responsive service recovery mechanisms The implementation of this customer-centric model led to improved delivery success rates, higher levels of customer satisfaction, and stronger customer loyalty, as buyers experienced more control, transparency, and trust in the delivery process.

[9]. **Soh, K.L. et al**. In general, logistics is often approached from two perspectives. Firstly, logistics is the science of organizing and managing the distribution of goods and services from the pre-production stage until the goods reach the consumer. investigated the inefficiencies of existing e-commerce shipment delivery systems. proposed a collaborative logistics information system that integrates real-time data sharing across all stakeholders in the supply chain, including suppliers, carriers, warehouses, and customers. resulted in improved delivery accuracy, faster response to delivery disruptions, enhanced visibility for customers, and ultimately, higher customer satisfaction due to better transparency and reliability throughout the delivery process.

[10] **UN ESCAP**. logistics is the management of the process of moving materials from warehousing to manufacturing into the final product to distribute to the consumers per their request. Secondly, logistics is also approached from the perspective of a service associated with the distribution and circulation process. In Vietnam, the official concept of logistics services was introduced for the first time in the 2005 Commercial Law of Vietnam (Article 233). “Logistic services are commercial activities whereby traders organize the performance of one or many jobs including reception, transportation, warehousing, yard storage of cargoes,

completion of customs procedures and other formalities and paperwork, provision of consultancy to customers, services of packaging, marking, delivery of goods, or other services related to goods according to agreements with customers in order to enjoy service charges".

[11]. **Quirk P Forder et al (2003)** examined the shortcomings of existing e-commerce shipment delivery systems, which often operated on rigid, standardized delivery frameworks that failed to consider the variability in customer expectations, geographic dispersion, and order complexities proposed a customer-adaptive logistics model that emphasized personalization, real-time communication, and adaptive resource allocation. Their model included advanced customer profiling, flexible scheduling based on customer availability, dynamic route optimization, and integrated feedback systems to monitor service quality continuously.

[12] **Han-ping et al** points out the fundamental differences between traditional logistics and e-logistics : while the distribution model of traditional logistics is "supply-driven push", e-logistics is "demand-driven pull", which shows that e-logistics is more flexible and responds better to market needs. E proposed a service-oriented logistics model that integrates customer segmentation, flexible delivery options, and real-time decision-making tools proposed a service-oriented logistics model that integrates customer segmentation, flexible delivery options, and real-time decision-making tools.

[13] **Yazdanifard**, **et al** (2011) it is observed that delivery is the most important part of ecommerce and the future of ecommerce relies on improved and satisfactory delivery services. Customers have become increasingly reluctant to order products. Existing System - Common Challenges Leading to Customer Dissatisfaction Delivery Delays: This is consistently cited as a major source of frustration. Customers expect timely delivery, and any deviation from promised timelines leads to negative experiences. Based on the research, a proposed system aimed at enhancing e-commerce shipment delivery customer satisfaction would likely focus on Real-time and Accurate .The results of the analysis would reveal which elements of the delivery system contribute to customer satisfaction and where improvements are needed. This information can be used to inform decisions about optimizing the existing system or implementing a new one.

[14] **Huang et al** (2006) has studied how ecommerce is affected by various factors in grocery delivery services. This study analysed four situational factors that influence customers to either prefer buying groceries online or from the store. It was found that people prefer that the groceries be delivered on time and this was the most influential factor that pushed the customers towards the online shopping portal. Existing System Common Challenges Delivery delays, lack of transparency, damaged or incorrect products, high delivery fees, inflexible delivery options, and inefficient customer service. Impact on Customer Satisfaction: Negative experiences lead to dissatisfaction, complaints, and customer churn. Expected Results Increased Customer Satisfaction: Improved delivery experiences lead to higher satisfaction

rates. Higher Customer Retention: Satisfied customers are more likely to make repeat purchases.

[15] **Sourabh Bhattacharya et al** (2015) found that e commerce is showing tremendous growth rate fuelled by boom in telecom and internet connections. Industries like travelling, banking, media, advertising, retailing have utilized the benefits being provided by E-commerce. Existing System Challenges Delivery Delays Timely delivery is crucial, with customers expecting next-day or same-day delivery options. Delays can lead to cart abandonment, negative reviews, and customer churn. Proposed System Improvements Real-time Tracking: Providing accurate and timely updates on shipment status. Faster Delivery Times Optimizing logistics and supply chain processes. Flexible Delivery Options Offering scheduled delivery slots, pick-up points, and diverse payment methods. Improved Packaging: Minimizing product damage during transit Expected Results Research suggests that factors like product quality, information quality, and customer service significantly impact customer satisfaction in e-commerce logistics

[16] **Averbakh et al** (2012) it is stated that around 50% of the people have purchased or services online, and around 28% of the online purchases have actually failed and around one out of five customers had bad experience over online purchases. Existing System Challenges Delays can lead to cart abandonment, negative reviews, and customer churn. Lack of Transparency/Tracking Inadequate tracking information increases anxiety and dissatisfaction. Real-time updates on order status and estimated arrival times foster trust and loyalty. Proposed System Improvements Real-time Tracking Providing accurate and timely updates on shipment status. Faster Delivery Times Optimizing logistics and supply chain processes. Expected Results Increased Customer Satisfaction Improved delivery experiences lead to higher satisfaction rates. Higher Customer Retention Satisfied customers are more likely to make repeat purchases.

[17] **Zeithaml et al** (2002) have analysed various sources of literature to conclude that the delivery service of any ecommerce business is one of the most crucial aspects to its success, in comparison to web-based presence and low prices. This aspect of ecommerce can only be improved upon analysis of customer perception of online service. Existing System Challenges Service Quality Gap Discrepancies between customer expectations and actual service delivery. Lack of Transparency Insufficient information about shipment status and delivery times. Proposed System Improvements Service Quality Enhancement Implementing processes to meet or exceed customer expectations. Expected Results Increased Customer Satisfaction: Improved service quality and transparency lead to higher satisfaction rates. Competitive Advantage Superior service quality differentiates the e-commerce platform from competitors.

[18] **Sabou et al** (2017) there has been a study on the behaviour of consumers in ecommerce which resulted in the findings that there are a group of customers who show deterrent behaviour post an unsatisfactory delivery experience. Existing System Unpredictable Delivery Time Shipment delivery times were highly variable, causing dissatisfaction Customers often could not track orders in real-time, leading to anxiety and lack of transparency Proposed System. Integration of logistics data with e-commerce platforms for seamless communication. Use of AI and data analytics for demand forecasting Real-Time Shipment

Tracking Implement GPS and IoT-based tracking systems. Provide customers with real-time updates and ETA predictions. Results general it was found that despite a group of people avoiding online shopping due to trust issues the number of new customers continues to grow at a rapid pace and it seems to show that the benefits of online shopping are better than its drawbacks.

[19] **Xanthidis Nicholas et al** (2007) have assessed the preferences of customers in Greece with regard to ecommerce. existing e-commerce shipment delivery system was characterized by fragmented logistics operations, limited integration between e-commerce platforms and third-party logistics providers, and a lack of end-to-end visibility for both companies and customers. The proposed system emphasized the integration of logistics management with e-commerce operations, introducing centralized information systems that allowed real-time tracking, better coordination among supply chain participants, and proactive communication with customers. As a result of these changes, companies experienced improved delivery reliability, increased customer satisfaction due to better communication and transparency, reduced operational inefficiencies, and stronger customer loyalty

.

[20] **Punakivi Sarannen et al** (2001) have explored the differences between manned and unmanned e-commerce deliveries for groceries in their work. the existing e-commerce shipment delivery systems primarily relied on traditional delivery models that were not optimized for the growing demands of online shopping. These systems faced issues such as inefficient last-mile delivery, high delivery costs, long waiting times for customers. The proposed system introduced the concept of *time-window based deliveries* and *dynamic delivery scheduling*, where customers could select preferred delivery slots, and companies could optimize delivery routes accordingly. The results demonstrated that by adopting these strategies, companies could significantly reduce delivery failures, improve customer convenience and satisfaction, optimize delivery efficiency, and lower operational costs.

[21] **Choi Lee et al** (2003) regarding consumer behaviour in ecommerce, it was found that USA consumers were relatively less influenced by risk factors and especially did not worry about customer service at all. the existing e-commerce shipment delivery systems struggled with fragmented supply chains, lack of coordination between retailers and logistics providers, and limited visibility across the entire delivery process. The proposed system focused on implementing Collaborative Planning, Forecasting, and Replenishment (CPFR**)** and integrating supply chain partners through advanced information systems. This approach enabled real-time data sharing, improved demand forecasting, better inventory management, and more accurate delivery scheduling. As a result, companies achieved greater synchronization across the supply chain, leading to shorter delivery lead times, improved order accuracy, and enhanced customer satisfaction.

[22] **Chang Wang** et al (2012) were able to identify that consumers expected four major mediums for these services: compensation, response speed, apologies, and contact channels. the existing e-commerce shipment delivery systems faced significant challenges such as inefficient inventory management, poor coordination between warehouses and

transportation providers, and limited responsiveness to sudden changes in customer demand. proposed system introduced the use of cloud-based supply chain management and data-driven decision-making to enhance flexibility and responsiveness. By leveraging real-time data analytics, companies could monitor inventory levels, optimize transportation routes, and quickly adapt to fluctuations in demand. As a result, companies experienced better inventory control, reduced delivery lead times, higher delivery accuracy, and improved customer satisfaction.

[23] **McKinnon et al** (2003) unattended delivery to homes and its security threats have been discussed. existing e-commerce shipment delivery systems faced major inefficiencies, particularly in the last-mile delivery stage, which led to high delivery costs, frequent delivery failures, and reduced customer satisfaction. The proposed system focused on improving **last-mile logistics** through the introduction of urban consolidation centres, better delivery scheduling, and alternative delivery points (such as parcel lockers and pickup stations**)**. These solutions aimed to reduce delivery distances, optimize vehicle utilization, and provide customers with more convenient delivery options. As a result, companies were able to lower delivery costs, improve delivery reliability, reduce environmental impact, and enhance overall customer satisfaction by offering flexible and timely delivery services.

[24] **Liao et al** (2011). the existing e-commerce shipment delivery systems were hindered by weak supply chain coordination, delayed information sharing, and limited visibility into shipment status. These inefficiencies often resulted in delayed deliveries, order inaccuracies, and frustrated customers due to the uncertainty surrounding their orders. The proposed system emphasized the adoption of integrated information systems and supply chain visibility platforms that allowed real-time data exchange between suppliers, logistics providers, and e-commerce platforms. Through technologies such as RFID, GPS tracking, and electronic data interchange (EDI), companies could monitor shipments in real time, predict potential disruptions, and proactively communicate with customers. As a result, companies achieved better shipment accuracy, faster response to delays, improved inventory management, and higher customer satisfaction due to transparent and reliable delivery processes.

[25] **Park Regan et al** (2004) the existing e-commerce shipment delivery systems were largely based on conventional transportation and distribution models that struggled to cope with the increasing complexity and volume of online orders. These systems lacked real-time coordination between e-commerce platforms, logistics providers, and customers, leading to delivery delays, inefficient routing, and limited flexibility in meeting customer preferences. The proposed system introduced the concept of Intelligent Transportation Systems (ITS) and logistics information integration, where advanced communication technologies, real-time traffic data, and dynamic route optimization tools were used to improve delivery efficiency. As a result, the implementation led to improved delivery accuracy, reduced transportation costs, faster delivery times, and higher levels of customer satisfaction

[26] **Chanana & Goele et al** (2012) discussed about the factors affecting the growth of e commerce in emerging markets like India. the existing e-commerce shipment delivery system was marked by operational inefficiencies such as poor coordination between sellers and

logistics providers, unpredictable delivery timelines, limited customer involvement in delivery decisions, and weak post-delivery support. The proposed system emphasized the adoption of customer-centric logistics models, where technology-driven solutions like real-time tracking, automated warehousing, advanced CRM systems, and flexible delivery scheduling were integrated into the e-commerce delivery process. As a result, businesses experienced improved on-time delivery rates, reduced operational costs, enhanced customer trust, and increased customer satisfaction due to better communication, transparency, and personalized delivery experiences.

[27] **Kaur Joshi et al** (2012) discuss the structure of electronic commerce, the advantages of it along with the challenges. the existing e-commerce shipment delivery systems were hindered by limited supply chain visibility, unoptimized last-mile delivery, poor demand forecasting, and insufficient coordination between e-commerce platforms and logistics service providers. These inefficiencies led to frequent delivery delays, inaccurate order fulfilment, higher logistics costs, and growing customer dissatisfaction. The proposed system focused on implementing technology-driven supply chain integration, leveraging tools such as predictive analytics, AI-powered demand forecasting, real-time shipment tracking, and improved warehouse automation. As a result, companies witnessed improvements in delivery speed, order accuracy, inventory management, and overall customer satisfaction.

[28] **Lee Park** (2010) talks about service failure refers to a service performance that falls below a customer's expectations. the existing e-commerce shipment delivery systems suffered from fragmented logistics networks, inefficient coordination between retailers and delivery providers, and a lack of real-time visibility throughout the supply chain. The proposed system introduced collaborative logistics models supported by real-time information sharing, integrated supply chain platforms, and dynamic delivery management systems. This approach allowed for better coordination among all supply chain participants, optimized delivery routes based on real-time traffic data, and improved inventory accuracy. As a result, companies achieved faster and more reliable deliveries, reduced logistics costs, and significantly improved customer satisfaction by offering greater convenience and transparency in the delivery process.

[29] **Hansen et al**. (2010) the existing e-commerce shipment delivery systems were characterized by limited responsiveness to customer demands, insufficient integration between inventory management and delivery operations, and poor visibility across the logistics network. The proposed system focused on implementing agile supply chain strategies, which emphasized flexibility, real-time data sharing, and stronger collaboration between e-commerce platforms, suppliers, and logistics providers. Technologies such as RFID, advanced inventory tracking, demand forecasting tools, and real-time shipment monitoring were integrated to allow faster decision-making and proactive issue resolution. As a result, companies experienced more accurate order fulfilment, reduced lead times, better inventory control, and enhanced customer satisfaction due to consistent and reliable deliveries.

[30] **Childers et al** (1987) found that dissatisfied customers spread negative WOM (word of mouth). experience. the existing delivery systems were primarily designed for traditional retail and catalog-based shopping, where delivery was slow, lacked flexibility, and

offered limited options for customer control over shipment timing and location. The proposed approach, which later became highly relevant for e-commerce, emphasized the importance of consumer convenience, flexible delivery models, and customer control over the delivery process. By offering delivery options such as home delivery, scheduled delivery windows, pickup points, and real-time tracking, the system could better align with customer preferences. The results demonstrated that when consumers were given more control, transparency, and convenience in the delivery process, overall satisfaction increased significantly.

[31] **Averbakh & Baysan et al** (2012) the existing e-commerce shipment delivery systems struggled with inefficient delivery route planning, high transportation costs, and limited ability to handle complex delivery scenarios involving multiple time windows and customer constraints. The proposed system introduced the use of advanced optimization algorithms and mathematical models, particularly focusing on the vehicle routing problem with time windows (VRPTW), to better manage delivery schedules, routes, and customer time preferences. By applying these advanced computational methods, companies could optimize delivery sequences, minimize travel distances, and ensure on-time deliveries even under complex constraints. As a result, businesses saw improved delivery efficiency, lower costs, better resource utilization, and higher customer satisfaction through more reliable and flexible delivery services.

[32] **Delarosa's et al** (2003) the existing e-commerce shipment delivery systems faced key challenges such as poor inventory visibility, lack of coordination between supply chain partners, inefficient order processing, and limited ability to respond to changing customer demands. The proposed system emphasized the implementation of collaborative supply chain management (CSCM) and real-time inventory tracking systems, allowing all stakeholders to share accurate and timely information. Technologies such as electronic data interchange (EDI), RFID, and integrated order management platforms were suggested to streamline communication, optimize inventory levels, and speed up order fulfilment. As a result, companies experienced fewer stockouts, faster deliveries, better responsiveness to customer needs, and improved customer satisfaction through reliable and consistent service.

[33] **Bellamkonda et al** (2014) found that the service quality of online shops is validating the tool in an Indian context. the existing e-commerce shipment delivery systems struggled with several challenges including delayed shipments, limited last-mile delivery options, inefficient tracking systems, and inadequate customer service support. he proposed system focused on enhancing last-mile logistics efficiency, implementing real-time tracking technologies, and adopting customer relationship management (CRM) systems to improve communication and service personalization. The model also recommended the use of automated warehouses, better demand forecasting, and flexible delivery time slots to reduce operational bottlenecks. As a result, companies observed improvements in on-time deliveries, enhanced customer communication, lower operational costs, and significantly higher levels of customer satisfaction.

[34] **Achuthan et al** (2011) expressed that certain factors that can build continued intention to use the same website in an online buying situation. The existing e-commerce shipment delivery systems were often limited by fragmented supply chains, inefficient coordination between logistics providers, and poor visibility across multiple stages of delivery. The proposed system introduced an integrated framework utilizing real-time information sharing, predictive analytics, and supply chain coordination platforms to enhance operational efficiency. Technologies like RFID, GPS tracking, data analytics, and machine learning algorithms were suggested to optimize inventory management, improve demand forecasting, and provide customers with real-time delivery updates. The results suggest that perceived usefulness and trust features of a website have appositive impact on continued intention factor in electronic buying.

[35] **Eid et al** (2011) the existing e-commerce shipment delivery systems faced issues such as inconsistent delivery performance, lack of trust in online transactions, limited customer interaction during the delivery process, and inadequate service quality. These problems negatively impacted customer satisfaction, especially in emerging e-commerce markets where consumer confidence was still developing. The proposed system emphasized building customer trust and satisfaction by integrating reliable delivery services, transparent communication, and advanced tracking technologies. The model also recommended incorporating secure payment systems, responsive customer service, and personalized delivery options to enhance the overall customer experience. As a result, businesses experienced improved customer loyalty, higher levels of satisfaction, reduced delivery complaints, and greater confidence in using e-commerce platforms.

[36] **Urvashi Tandon et al** (2017) the existing e-commerce shipment delivery systems were found to suffer from various issues such as delayed deliveries, limited flexibility in delivery options, inefficient return processes, and poor communication with customers, all of which contributed to lower levels of customer satisfaction. The proposed system emphasized the importance of adopting customer-centric logistics models that focus on enhancing service quality through real-time tracking, proactive communication, efficient return handling, flexible delivery scheduling, and personalized delivery services. The integration of advanced technologies such as GPS, CRM tools, and data analytics allowed businesses to better understand customer preferences and deliver more tailored services. As a result, companies observed improved delivery reliability, faster problem resolution, enhanced customer engagement, and higher customer satisfaction and loyalty.

[37] **Desai et al** (2016) the existing e-commerce shipment delivery systems faced significant challenges such as delayed deliveries, inefficient last-mile logistics, lack of real-time order visibility, and limited customer interaction during the delivery process. These issues often led to customer dissatisfaction due to unmet expectations regarding delivery speed, accuracy, and convenience. The proposed system emphasized the integration of technology-enabled logistics solutions such as real-time shipment tracking, automated warehouse management, predictive analytics for demand forecasting, and optimized route planning. Furthermore, the system encouraged enhanced customer engagement through timely updates, flexible delivery options, and efficient return management processes. As a result of

implementing these improvements, businesses experienced higher delivery accuracy, reduced operational costs, faster order fulfilment, and significantly improved customer satisfaction and loyalty.

[38] **Morganti et al** (2014) the existing e-commerce shipment delivery systems were mainly challenged by inefficiencies in urban last-mile logistics, including traffic congestion, limited delivery windows, high failed delivery rates, and environmental concerns due to increasing delivery vehicle emissions. The proposed system focused on implementing sustainable urban logistics solutions, such as urban consolidation centers, alternative delivery methods like cargo bikes and electric vehicles, parcel lockers, and improved delivery time coordination with customers. These innovations aimed to reduce congestion, optimize delivery routes, lower emissions, and ensure timely and successful deliveries. As a result, companies observed reduced delivery costs, improved punctuality, lower environmental impact, and higher levels of customer satisfaction due to greater delivery flexibility and reliability.

[39] **Lawal et al** (2015) he existing e-commerce shipment delivery systems faced problems such as inefficient order processing, inadequate transportation infrastructure, unreliable delivery timelines, and poor coordination between logistics partners. The proposed system focused on the integration of ICT (Information and Communication Technology) tools, including automated order processing, real-time shipment tracking, advanced inventory management, and stronger collaboration among supply chain partners. Additionally, the system encouraged better customer service practices such as proactive communication, flexible delivery options, and improved complaint resolution mechanisms. As a result of implementing these strategies, companies were able to achieve faster order fulfilment, improved delivery accuracy, reduced operational costs, and higher levels of customer satisfaction and loyalty.

[40] **Wilkins l et** al the existing e-commerce shipment delivery systems often suffer from multiple inefficiencies such as delayed dispatches, inaccurate tracking information, inadequate last-mile delivery coordination, and limited responsiveness to customer grievances. proposed a data-driven, integrated logistics management system that leverages real-time tracking technologies, predictive analytics for demand forecasting, and AI-powered customer communication interfaces. The proposed system emphasizes proactive shipment monitoring, dynamic route optimization, and instant customer updates regarding delivery status. Result E-commerce success in virtual environments depended on technical skills and operational competencies. Cloud-based systems and collaboration tools improved service scalability.

[41] **Gnaneshwar et al** (2013) the existing e-commerce shipment delivery systems were challenged by manual inventory management, inefficient route planning, limited order visibility, and weak real-time coordination between supply chain stakeholders. These limitations resulted in frequent order delays, stockouts, higher logistics costs, and ultimately, reduced customer satisfaction due to unreliable service. The proposed system introduced automation in warehousing, intelligent transportation systems, and integrated supply chain management platforms to address these issues. Technologies such as RFID, GPS tracking, predictive analytics, and real-time inventory control were implemented to enable better forecasting, efficient resource utilization, and proactive issue resolution. Furthermore,

improved customer communication channels, such as order tracking portals and timely delivery notifications, enhanced the overall customer experience. As a result, businesses saw significant improvements in delivery speed, order accuracy, inventory turnover, and customer satisfaction levels.

[42] **Zhang & Song et al** (2016) a study was conducted to evaluate the customer's perception of delivery speed as a differentiating factor for ecommerce sales in China. the existing e-commerce shipment delivery systems were hampered by poor demand forecasting, insufficient capacity planning, inefficient last-mile delivery, and limited flexibility in handling dynamic customer requirements. The proposed system introduced the use of big data analytics, machine learning algorithms, and dynamic routing optimization to better predict customer demand, optimize inventory distribution, and improve delivery scheduling. Advanced technologies like real-time traffic monitoring, intelligent warehouse management, and customer behaviour analysis were integrated into the logistics network to create a more adaptive and responsive delivery system. As a result, businesses experienced improved order fulfilment accuracy, reduced delivery lead times, enhanced delivery success rates, and significantly higher customer satisfaction.

[43] **Mingyao et al** (2015) found that online customer complaint behaviour influences not only customer loyalty, but also other customers purchase intentions, it is important to examine the antecedents of customer complaint behaviour. The outcomes from this information show that signal credibility impacts online client objection goals. Moreover, signal credibility can direct the connection between consumer loyalty with post-buy administrations and online client objection goals.

[44] **Simona et al** (2017) the existing e-commerce shipment delivery systems were facing challenges such as lack of real-time visibility, poor last-mile delivery coordination, limited personalization of delivery options, and inefficient return processes. The proposed system emphasized a customer-centric logistics model that incorporated real-time tracking technologies, advanced customer relationship management (CRM) systems, predictive analytics, and flexible last-mile delivery solutions. By offering features like personalized delivery time slots, interactive delivery rescheduling, and seamless return handling, the system was able to align more closely with individual customer preferences and expectations. As a result, businesses achieved higher delivery reliability, improved operational efficiency, faster problem resolution, and significantly enhanced customer satisfaction and loyalty.

[45] Rutner et al., 2003 the existing e-commerce shipment delivery systems were primarily limited by fragmented logistics networks, inefficient inventory management, limited coordination between supply chain partners, and high uncertainty in delivery timelines. The proposed system recommended the development of an integrated supply chain management approach, where collaborative partnerships, shared information systems, and real-time data exchange were implemented to improve coordination and visibility across the entire delivery process. The study also emphasized the use of information technology tools such as electronic data interchange (EDI), warehouse management systems (WMS), transportation management systems (TMS), and demand forecasting models to enhance accuracy and responsiveness. As

a result, companies observed improved delivery reliability, better inventory control, faster response to customer demands, reduced logistics costs, and significantly enhanced customer satisfaction.

[46] **Whang et al**, (2001) the existing e-commerce shipment delivery systems were challenged by lack of supply chain coordination, limited information sharing between stakeholders, poor demand forecasting, and inefficient inventory management. These inefficiencies often resulted in delivery delays, stockouts, higher operational costs, and ultimately, lower customer satisfaction. The proposed system emphasized the adoption of collaborative supply chain management (CSCM), where real-time data sharing, coordinated planning, and integrated information systems allowed for better synchronization among suppliers, retailers, and logistics providers. The use of technologies such as electronic data interchange (EDI), advanced forecasting models, and inventory visibility tools improved both supply chain responsiveness and delivery reliability. As a result, businesses achieved more accurate order fulfilment, reduced inventory holding costs, minimized delivery delays, and improved customer satisfaction by providing more consistent and transparent service.

[47] Chosin et al (2017) The existing e-commerce shipment delivery systems faced several critical issues, including limited delivery infrastructure, poor coordination between logistics partners, inconsistent delivery times, and lack of real-time order tracking. These inefficiencies led to delays, high operational costs, and declining customer trust and satisfaction. The proposed system focused on enhancing e-commerce logistics by integrating modern information technology solutions, real-time tracking systems, improved last-mile delivery strategies, and advanced inventory management tools. The system also emphasized the importance of customer engagement, allowing customers to personalize delivery times and receive proactive updates throughout the delivery process. As a result of these improvements, businesses experienced better delivery performance, faster and more accurate order fulfilment, improved resource utilization, and significantly higher customer satisfaction and loyalty.

[48] **Reicher et al (**2015) the existing e-commerce shipment delivery systems were characterized by fragmented logistics operations, limited real-time visibility, inefficient inventory distribution, and a lack of flexibility in responding to dynamic customer demands. These weaknesses frequently resulted in delayed shipments, inaccurate delivery estimates, and growing customer dissatisfaction. The proposed system focused on the adoption of integrated logistics platforms, incorporating real-time data analytics, centralized inventory management, dynamic routing algorithms, and advanced forecasting techniques to optimize delivery operations. Additionally, the system promoted flexible delivery windows, enhanced communication with customers, and proactive issue resolution mechanisms to improve the overall customer experience. As a result, businesses achieved greater operational efficiency, faster and more reliable deliveries, better inventory turnover, and significantly improved customer satisfaction and loyalty

[49] **Fernandes et al.,** (2018) The existing e-commerce shipment delivery systems struggled with delayed deliveries, insufficient last-mile delivery solutions, lack of real-time tracking, and poor customer service integration. These inefficiencies often led to unmet

customer expectations, failed deliveries, and increased dissatisfaction, especially in competitive e-commerce environments where speed and reliability are critical. The proposed system focused on introducing technology-driven solutions such as automated warehouse systems, real-time GPS tracking, optimized delivery route algorithms, and integrated customer service platforms. Additionally, the system emphasized personalized delivery options, proactive communication, and efficient reverse logistics processes to enhance the overall customer experience. As a result of implementing these innovations, companies experienced reduced delivery times, higher delivery accuracy, improved operational efficiency, and significantly increased customer satisfaction and retention.

[50] **Hu Zhao et al** (2018) the existing e-commerce shipment delivery systems were challenged by inefficient logistics networks, lack of integration between suppliers and carriers, limited real-time visibility, and poor adaptability to fluctuating customer demands. The proposed system introduced the integration of smart logistics technologies, including Internet of Things (IoT) sensors, big data analytics, cloud-based supply chain management platforms, and AI-powered demand forecasting models. These technologies enabled real-time monitoring, dynamic inventory allocation, predictive maintenance for delivery fleets, and adaptive routing strategies. Furthermore, by enhancing customer interaction through personalized delivery scheduling and instant notifications, the system significantly improved the service quality. As a result, companies achieved more efficient delivery operations, lower costs, better delivery accuracy, faster response to customer needs, and higher customer satisfaction levels.

[51**] Van Der Vorst et al**. (2009) the existing e-commerce shipment delivery systems faced major limitations such as lack of supply chain visibility, inefficient coordination among partners, delays in information sharing, and poor responsiveness to demand variability. The proposed system introduced the concept of supply chain redesign based on real-time information sharing, process integration, and collaborative planning. Technologies like supply chain event management (SCEM), integrated IT platforms, and advanced forecasting tools were incorporated to enhance transparency, improve coordination, and enable real-time decision-making across the supply chain. As a result, companies experienced reduced inventory costs, improved delivery reliability, faster response times, and significantly higher customer satisfaction.

[52**] Anderson et al.** (1998**)** the existing e-commerce shipment delivery systems were limited by fragmented supply chain processes, slow information flow, minimal integration between suppliers and distributors, and rigid delivery schedules. The proposed system emphasized greater supply chain integration, the use of information technologies for real-time data sharing, and flexible logistics networks that could adapt to fluctuating customer demands. By implementing enterprise resource planning (ERP) systems, electronic data interchange (EDI), and collaborative planning tools, companies improved coordination across the supply chain and gained better visibility into inventory and shipment status. As a result, businesses experienced lower inventory holding costs, improved delivery reliability, faster order processing, and significantly enhanced customer satisfaction and loyalty.

[53] **Kafel and Sikora, et al 2014;** The existing e-commerce shipment delivery systems were hindered by poor quality management in logistics operations, weak monitoring of delivery processes, inconsistent service quality, and limited focus on continuous improvement The proposed system emphasized the application of Total Quality Management (TQM) principles and continuous improvement frameworks within the e-commerce logistics sector. This included setting clear quality standards, regular performance monitoring, customer feedback integration, employee training, and process optimization. Additionally, the use of Key Performance Indicators (KPIs) and process audits helped companies track and improve delivery reliability and customer service levels. As a result, organizations experienced better delivery accuracy, reduced operational errors, enhanced service consistency, and significant improvements in customer satisfaction and loyalty.

[54] **Islam et al**., (2011)**;** Customer satisfaction can be defined as a state in which his expectations match his perception of the actual service received. the existing e-commerce shipment delivery systems faced major challenges such as inefficient transportation networks, limited last-mile delivery capacity, inadequate use of technology, and lack of real-time shipment visibility. The proposed system focused on improving e-commerce delivery performance through the adoption of advanced information and communication technologies (ICTs), real-time tracking systems, optimized route planning, and the integration of transportation management systems (TMS). As a result of these improvements, companies experienced faster delivery times, reduced delivery failures, improved cost-efficiency, and higher levels of customer satisfaction and retention.

[55**] Grigoroudis and Siskos et al** (2010) Customer satisfaction measurement is one of the most important issues concerning business organizations, the existing e-commerce shipment delivery systems were limited by insufficient focus on customer satisfaction metrics, lack of systematic service quality evaluation, inconsistent delivery performance, and poor responsiveness to customer needs. The proposed system applied multi-criteria decision analysis (MCDA) and customer satisfaction evaluation frameworks to systematically assess and improve the quality of shipment delivery services. By systematically integrating customer expectations into service design and delivery operations, companies achieved more reliable deliveries, better alignment with customer preferences, proactive problem resolution, and significant improvements in customer satisfaction and retention.

[56] **Domingues et al**., 2015 Effectiveness of a company depends largely on the proper identification of the customer as well as his understanding of the expectations and impact on the market. The existing e-commerce shipment delivery systems were affected by inefficient performance measurement, lack of standardized logistics performance indicators, limited real-time monitoring, and poor alignment between logistics operations and customer expectations. The proposed system focused on developing a comprehensive logistics performance measurement framework that included key performance indicators (KPIs), customer feedback integration, continuous monitoring, and real-time data analysis. As a result, businesses experienced more consistent and timely deliveries, reduced operational inefficiencies, and significant improvements in overall customer satisfaction and loyalty.

[57] **Fynes et al.,** 2005 Consumer behaviour in the market depends on many factors and relatively complicated processes. the existing e-commerce shipment delivery systems struggled with weak supply chain relationships, lack of trust among supply chain partners, poor communication, and limited collaboration. The proposed system emphasized building supply chain relationship quality (SCRQ) by fostering trust, commitment, information sharing, and cooperation among all supply chain stakeholders. Through stronger partnerships and open communication, companies were able to improve coordination, responsiveness, and flexibility in their delivery processes. As a result, businesses experienced enhanced delivery reliability, reduced lead times, better inventory management, and significant improvements in customer satisfaction and loyalty.

[58] **Gerson**, **et al** (1993) According to these regularities, customer satisfaction should be measured and translated into a number of measurable parameters. the existing e-commerce shipment delivery systems (or early forms of remote ordering systems) were often characterized by limited customer feedback mechanisms, lack of structured customer service processes, and minimal use of data to improve service quality. he proposed system emphasized the integration of customer satisfaction measurement frameworks, where organizations would systematically collect and analyze customer feedback, monitor service performance, and address delivery issues proactively. As a result, companies experienced improved customer service responsiveness, better resolution of delivery issues, more consistent service quality, and stronger customer loyalty.

[59] **Manusamy et al 2008** Customer satisfaction is measured by Servqual Model, Expectancy–Disconfirmation Model, Performance Only Model (Servperf), Attribute Importance Model, Attribution Model, Affective Model, the existing e-commerce shipment delivery systems faced challenges such as limited focus on customer relationship management, poor integration between logistics and customer service departments, and insufficient use of technology to personalize the delivery experience. The proposed system introduced a customer-centric logistics model that integrated Customer Relationship Management (CRM) systems with logistics operations to better understand and respond to customer preferences. By utilizing customer data analytics, real-time tracking, personalized communication, and service recovery mechanisms, companies were able to enhance both operational efficiency and customer experience. As a result, businesses saw reduced delivery issues, quicker problem resolution, improved service personalization, and significantly higher customer satisfaction and loyalty.

[60] **Wolniak et al** 2012 The Servqual method is based on measuring the differences between the level of meeting the expectations and the perception of the services by a customer. n the existing e-commerce shipment delivery systems, many companies primarily focus on order processing and dispatch, often neglecting the broader aspects of customer experience such as timely delivery, communication, and service recovery the proposed system integrates real-time tracking, predictive analytics for delivery times, and proactive communication strategies to keep customers informed throughout the shipment process. It also includes advanced feedback mechanisms to promptly resolve customer complaints and improve service

quality continuously. The results of implementing the proposed system indicate a substantial improvement in overall customer satisfaction.

[61] **Parasuraman et al.,** (1988) The starting point for the method development is the assumption that quality assessment. the existing e-commerce, many organizations struggle to meet the full spectrum of customer expectations. SERVQUAL dimensions: tangibles, reliability, responsiveness, assurance, and empathy. In many current systems, gaps exist particularly in reliability (consistent delivery performance) and responsiveness (quick resolution of issues), leading to customer dissatisfaction when deliveries are delayed, mishandled, or poorly communicated. the proposed system incorporates advanced logistics management, real-time shipment tracking, proactive customer notifications, and AI-powered customer service to address issues promptly of the quality of services performed by customers results from the discrepancy between its expectations and satisfaction with the services provided.

[62] **Franceschini et al** 2000 This is due to the fact that the quality experienced during the consumption of the service in a subconscious manner is compared to the expected quality, resulting in a quality conscious. The existing e-commerce shipment delivery systems often face challenges in effectively measuring and controlling the quality of logistics services. the proposed system incorporates a more structured performance measurement and management approach, integrating key performance indicators (KPIs) such as on-time delivery rate, order accuracy, and customer complaint resolution time. As a result of implementing the proposed system, on-time delivery rates increased by 35%, order accuracy improved by 25%, and customer complaint resolution times were reduced by 50%

[63] **Dellana et al** (2014) rightly noted, the quality of services perceived by clients is shaped during three processes: production, experience and quality assessment. In the existing e-commerce shipment delivery systems, companies often face variability in service quality due to fragmented processes, limited real-time monitoring, and inconsistent customer service.. he proposed system integrates Six Sigma principles to analyze and optimize critical processes in the supply chain. As a result of implementing the proposed system, delivery delays were reduced by 40%, order accuracy improved by 30%, and customer satisfaction ratings increased by 25%.

. [64] **Sampaio et al**. (2011) Creating optimal quality services does not stop at the production stage and does not focus only concern on the resources of service provider. the existing e-commerce shipment delivery systems were often characterized by fragmented logistics operations, limited coordination between supply chain actors, and inconsistent service quality, which negatively impacted customer satisfaction levels. It is necessary to interact with customers to develop high-quality services. proposed a structured quality management approach, integrating Total Quality Management (TQM) principles with logistics processes. Their proposed system emphasized continuous monitoring, process standardization, proactive issue resolution, and enhanced communication across the supply chain.

[65] **Zimon et al** (2017) Even a product with excellent features and characteristics will not fully satisfy the customer if the logistic processes related to its displacement in the supply chain

fail. The existing systems, as identified in their study, often lacked proper quality control mechanisms, resulting in frequent errors such as delayed deliveries, damaged goods, and poor communication with customers. proposed the integration of Total Quality Management (TQM) and Lean Management principles into logistics operations. As a result of implementing these strategies, companies observed a significant reduction in shipment errors and delivery delays, improved product handling, and enhanced customer communication.

[66] **Agatz et al**., (2008) investigated the inefficiencies of existing e-commerce shipment delivery systems, which were often characterized by rigid delivery schedules, limited customer choice in delivery options, and suboptimal routing decisions. proposed a dynamic and customer-centric delivery scheduling model that integrates real-time demand information with flexible delivery windows and optimized routing algorithms. Their model allowed customers to select preferred delivery times while enabling companies to balance delivery efficiency and service quality. The results demonstrated that the proposed system significantly improved delivery reliability, reduced operational costs, and enhanced overall customer satisfaction by offering more convenient and reliable delivery services.

[67] **Cho et al.,** 2008 Company's logistics capacity has significant and positive effects on logistics performance in E-commerce market. analyzed the shortcomings of existing e-commerce shipment delivery systems, which often failed to effectively manage reverse logistics, particularly in handling returns and exchanges. proposed an integrated reverse logistics model that streamlined the return process, improved coordination among logistics providers, and established clear guidelines for handling returned goods. proposed an integrated reverse logistics model that streamlined the return process, improved coordination among logistics providers, and established clear guidelines for handling returned goods.

[68**]** **Bask et al**., (2012) The modern logistics become the most important means to improve the efficiency of material flow, reducing distribution costs in various industries; studied the limitations of existing e-commerce shipment delivery systems, which primarily operated using standardized logistics models with limited customization based on customer needs. Large numbers of practices are carried out in the e-commerce logistics. The proposed system enabled greater adaptability and responsiveness to diverse customer demands. resulting in lower customer satisfaction, especially when expectations for personalized services were not met.

[69] **Jones Lang La Sall et al**, 2015 Forecasts done by PMR research anticipated an increase in spending on internet transactions until it reaches 10 per cent of retail sales in 2020. analyzed the challenges faced by existing e-commerce delivery systems, particularly the limitations in last-mile delivery infrastructure. To overcome these issues, the study proposed the development of urban logistics hubs and the strategic placement of fulfilment centers closer to high-demand areas. The results of implementing these solutions demonstrated significant improvements in delivery speed, reduced transportation costs, and enhanced customer satisfaction due to faster, more reliable, and transparent delivery services.

[70] **Vigil Joseph**, et al (2017) the direction of the research is to improve the bound and thus reducing the cost and time needed for delivery. examined the shortcomings of existing e-

commerce shipment delivery systems, which were often hindered by a lack of real-time visibility, inefficient coordination between logistics partners, and limited customer engagement during the delivery process. The proposed system aimed to provide both companies and customers with up-to-date information on shipment status, estimated delivery times, and proactive alerts for any disruptions. The implementation of these technologies resulted in improved operational efficiency, more accurate delivery scheduling, and higher customer satisfaction, as customers valued the transparency and reliability of the enhanced delivery process.

[71] **Zeithaml**, et al (2002) focused on the role of service quality in e-commerce shipment delivery and its impact on customer satisfaction. The existing systems at that time often emphasized transactional efficiency while neglecting critical service quality dimensions such as reliability, responsiveness, assurance, and empathy. proposed a service quality framework (SERVQUAL) tailored to the e-commerce logistics context, emphasizing the importance of meeting customer expectations across all service dimensions. By improving communication, ensuring timely deliveries, handling issues empathetically, and building customer trust, companies could enhance the perceived service quality. As a result, customers frequently experienced inconsistent delivery performance, poor communication, and lack of trust in the delivery process.

[72] **Hübner, Kuhn, et al** (2016) studied the inefficiencies of existing multi-channel and omnichannel e-commerce shipment delivery systems, which often suffered from fragmented operations between online and offline channels, leading to inconsistent delivery performance, inventory shortages, and higher operational costs. the authors proposed a fully integrated omnichannel logistics system that synchronized inventory management, order fulfilment, and delivery processes across all channels. This system included centralized inventory visibility, flexible fulfilment options. The results showed that companies implementing such integrated solutions achieved greater inventory accuracy, faster and more reliable deliveries, and significantly improved customer satisfaction by offering more convenient and consistent shopping and delivery experiences across all channels.

[73] **Xing, Grant, et al** (2011) The interface between retailers and logistics service providers in the online market analyzed the limitations of existing e-commerce shipment delivery systems, which often lacked coordination between retailers and logistics providers, resulting in delays, stockouts, and poor last-mile delivery performance. These inefficiencies frequently led to customer dissatisfaction due to unreliable delivery times, missing items, and inadequate tracking information. the authors proposed a collaborative logistics model that emphasized stronger partnerships between e-retailers and third-party logistics providers (3PLs), integrated information sharing, and real-time visibility across the supply chain. he results showed significant improvements in delivery reliability, inventory accuracy, and customer satisfaction, as customers experienced more timely deliveries, accurate order fulfilment, and better communication throughout the delivery process.

[74] **Punakivi, Yrjöla, et al** (2001) examined the limitations of traditional home delivery systems in e-commerce, which were generally built around fixed delivery routes and lacked

flexibility to accommodate varying customer needs. To overcome these issues, the authors proposed a dynamic delivery model that incorporated time-window-based home delivery and drop-point alternatives. Their model used demand forecasting, route optimization, and customer-preferred delivery time slots to reduce failed delivery attempts and operational costs. The results of their simulations and pilot studies showed that such a model improved delivery efficiency, reduced last-mile logistics expenses, and significantly increased customer satisfaction by offering more reliable and convenient delivery options tailored to customer preferences.

[75] **Sparks et al**. (2006) explored the limitations of existing e-commerce delivery systems, which often failed to address the emotional and psychological aspects of customer satisfaction related to delivery experiences. Existing systems primarily focused on operational efficiency but overlooked factors such as customer communication, service recovery after delivery failures, and managing customer expectations during delays. proposed a customer-centric service recovery model that emphasized proactive communication, prompt resolution of delivery problems, and empathetic customer service interactions during disruptions. The results indicated that companies adopting such customer-oriented service recovery practices experienced improved customer satisfaction, higher loyalty, and stronger long-term customer relationships, even in cases of delivery failures.

[76**]** **Fernie et al** (2014) Logistics and Retail Management: Emerging Issues and New Challenges in the Retail Supply Chain. Existing systems struggled to handle high order volumes, leading to frequent delivery delays, stockouts, and poor visibility for customers. proposed the adoption of agile logistics networks that incorporate advanced forecasting, dynamic inventory management, and innovative last-mile solutions such as local consolidation centers, crowd-sourced delivery, and micro-fulfilment hubs. The results demonstrated that companies implementing such agile and adaptive logistics systems achieved significant reductions in delivery times, improved inventory accuracy, and enhanced customer satisfaction due to faster, more reliable, and convenient delivery experiences.

[77] **Hübner, Holzapfel, et al** (2015) Distribution systems in omni-channel retailing. investigated the limitations of existing e-commerce fulfilment systems, particularly in multi-channel retail environments where retailers struggled to synchronize their online and offline inventory and delivery operations. Traditional systems were often siloed, leading to inventory inaccuracies, stockouts, delayed shipments, and inconsistent customer experiences across channels. To address these challenges, the authors proposed an integrated fulfillment strategy combining centralized and decentralized fulfilment centers with real-time inventory visibility across all channels. he results demonstrated that retailers adopting this integrated system achieved improved delivery speed, higher order accuracy, better inventory utilization, and enhanced customer satisfaction by providing more reliable, flexible, and convenient delivery services.

[78] **Zhou** et al (2007) examined the limitations of existing e-commerce shipment delivery systems, which often focused heavily on transaction completion while neglecting post-purchase logistics performance. proposed a holistic supply chain integration model that

emphasized collaboration between retailers, suppliers, and logistics providers through shared information systems, real-time order tracking, and joint demand forecasting. The results showed that companies adopting this integrated model experienced improved delivery performance, reduced stockouts, and significantly higher customer satisfaction due to more reliable service and better communication throughout the delivery process.

[79] **Boyer et al** (2005) Customer behavioural intentions for online purchases: An examination of fulfilment method and customer experience level. explored the weaknesses in existing e-commerce shipment delivery systems, which often provided limited delivery options, lacked real-time order visibility, and failed to incorporate customer preferences into the fulfillment process. Customers frequently faced rigid delivery schedules, delayed shipments, and poor communication, leading to dissatisfaction and reduced loyalty. proposed a customer-responsive delivery model that emphasized flexibility, transparency, and personalization.

[80] **Nguyen, T. H. et al** (2020) Factors affecting online shopping satisfaction in Vietnam: An empirical study. Journal of Asian Finance, Economics and Business. examined the weaknesses of existing e-commerce shipment delivery systems, which often suffered from poor demand forecasting, limited capacity planning, and weak coordination between suppliers and logistics providers. These inefficiencies frequently led to delayed deliveries, order cancellations, and increased operating costs, ultimately reducing customer satisfaction. proposed an integrated demand-driven logistics model that utilizes advanced analytics, real-time data sharing, and collaborative planning among supply chain partners. The implementation of this system resulted in improved order fulfilment rates, reduced lead times, and higher customer satisfaction, as customers experienced more timely, accurate, and reliable delivery services.

2.3 Research Gaps

While significant research exists on factors influencing e-commerce shipment delivery customer satisfaction (e.g., speed, reliability, tracking, cost), several critical gaps remain in the literature. A notable deficiency lies in the lack of standardized measurement instruments specifically designed to capture the nuanced customer experience within the "last-mile" delivery phase, making cross-study comparisons and robust model development challenging. Furthermore, there's insufficient exploration into the direct correlation between "logistics value" as perceived by customers and its tangible impact on online retailer performance, including comprehensive methodologies to quantify this value. The literature also presents a limited focus on how diverse customer segments (e.g., Gen Z, rural customers, business vs. individual consumers) perceive and prioritize various delivery aspects, requiring more granular investigations into their unique expectations, emotions, and behaviours. Lastly, while emerging technologies like AI and real-time tracking are discussed, more in-depth studies are needed to understand their precise influence on customer satisfaction and the associated new challenges or opportunities they present, particularly concerning novel delivery models such as drone or autonomous vehicle deliveries.

2.4 SUMMARY

The existing literature extensively highlights the critical role of shipment delivery in e-commerce customer satisfaction, identifying factors like delivery speed, reliability, tracking capabilities, and service quality as paramount. Studies consistently show that prompt and dependable shipping significantly influences customer behaviour, brand loyalty, and repeat purchases. Real-time tracking and advancements in logistics are recognized for reducing customer anxiety and enhancing trust. However, the survey reveals a need for more standardized measurement instruments to gauge the intricate customer experience during last-mile delivery. There's also a gap in understanding how "logistics value" directly translates into retailer performance and how diverse customer segments, with their unique expectations, perceive and prioritize different delivery aspects.

CHAPTER 3

EXISTING SYSTEM

3.1 Manual Customer Feedback Coding

Manual customer feedback coding involves systematically categorizing qualitative feedback to extract meaningful insights. This process typically starts with collecting feedback from various channels such as surveys, reviews, and support interactions. Each piece of feedback is then read and assigned codes or labels that represent specific themes or sentiments. For instance, comments about delivery delays might be coded under "logistics issues," while feedback praising prompt delivery could fall under "positive delivery experience." This method allows for the identification of recurring themes and patterns in customer feedback.

The coding process often employs a hierarchical structure, where broad categories are divided into subcategories for more granular analysis. For example, "product issues" might be subdivided into "packaging problems," "product defects," and "incorrect items." This structured approach facilitates a deeper understanding of specific areas affecting customer satisfaction. By quantifying the frequency of each code, businesses can prioritize areas that require immediate attention.

Manual coding is particularly useful for small to medium-sized enterprises (SMEs) that may not have access to advanced analytical tools. It provides a cost-effective way to analyze customer feedback and make data-driven decisions. However, this method can be time-consuming and may not be scalable for large volumes of data. Despite its limitations, manual coding remains a valuable tool for gaining qualitative insights into customer experiences.

3.2 Cognitive walkthroughs

Cognitive walkthroughs Cognitive walkthroughs are a usability evaluation method used to assess the user experience of a system or process. In the context of e-commerce shipment, this involves stepping through the entire delivery process from the customer's perspective to identify potential pain points. The evaluator considers each step a customer would take, such as placing an order, tracking the shipment, and receiving the package, asking questions like "Will the user know what to do at this step?" and "Will the user notice that the correct action is available?"

This method helps in identifying areas where customers might encounter confusion or frustration, such as unclear tracking information or lack of delivery updates. By simulating the customer's journey, businesses can proactively address issues that may negatively impact customer satisfaction. For example, if the walkthrough reveals that customers are not receiving timely notifications about their shipments, the company can implement automated alerts to keep customers informed.

Cognitive walkthroughs are particularly effective during the design and testing phases of e-commerce platforms. They allow businesses to refine their processes and interfaces before full-scale implementation, reducing the likelihood of customer dissatisfaction. While this method requires time and careful planning, the insights gained can lead to significant improvements in the customer experience.

3.3 Manual Shipment Tracking Logs

Manual shipment tracking involves maintaining detailed logs of each shipment's progress through various stages of the delivery process. This includes recording timestamps for order processing, dispatch, transit, and delivery. By analyzing these logs, businesses can identify patterns such as frequent delays in specific regions or with particular carriers. This information is crucial for pinpointing inefficiencies and implementing targeted improvements.

For example, if the logs reveal that shipments handled by a certain courier consistently experience delays, the company can investigate the cause and consider alternative carriers. Similarly, if delays are prevalent in a specific geographic area, the business might explore options like establishing a local distribution center to expedite deliveries. Manual tracking also allows for the monitoring of key performance indicators (KPIs) such as on-time delivery rates and average transit times, providing a clear picture of the logistics performance.

While manual tracking can be labor-intensive, especially for businesses with high shipment volumes, it offers a straightforward approach to monitoring and improving delivery operations. For SMEs, this method provides valuable insights without the need for sophisticated tracking systems. However, as the business grows, transitioning to automated tracking solutions may become necessary to handle the increased complexity and volume of data.

3.4 Problem statements of manual methods

- **Inconsistent Data Recording**: Manual methods often lead to discrepancies in data entry, making it challenging to maintain accurate records of shipments and customer feedback.
- **Limited Scalability**: As the volume of orders increases, manual analysis becomes time-consuming and prone to errors, hindering the ability to scale operations effectively.
- **Delayed Issue Resolution**: Without real-time data analysis, identifying and addressing delivery issues promptly becomes difficult, negatively impacting customer satisfaction.
- **Lack of Comprehensive Insights**: Manual analysis may overlook subtle patterns and correlations in data, limiting the depth of insights that can be derived.
- **Resource Intensive**: Manual tracking and analysis require significant human resources, diverting attention from other critical business functions.

CHAPTER 4
PROPOSED SYSTEM

4.1 Proposed Algorithm

This proposed algorithm introduces a unique combination of comprehensive exploratory data analysis (EDA) and unsupervised feature extraction using FastICA, enriched with diverse statistical visualizations not commonly fused in prior surveys. By integrating 30 different types of visual plots, including advanced KDEs, swarm plots, and ECDFs, alongside Independent Component Analysis on normalized key numeric variables like CSAT Score, Item Price, and Handling Time, the algorithm provides both visual interpretability and statistical feature independence. This method diverges from typical linear correlation or PCA-only pipelines by utilizing non-Gaussian independent component modeling, enabling deeper insight into latent customer satisfaction drivers that are often hidden in traditional analysis.

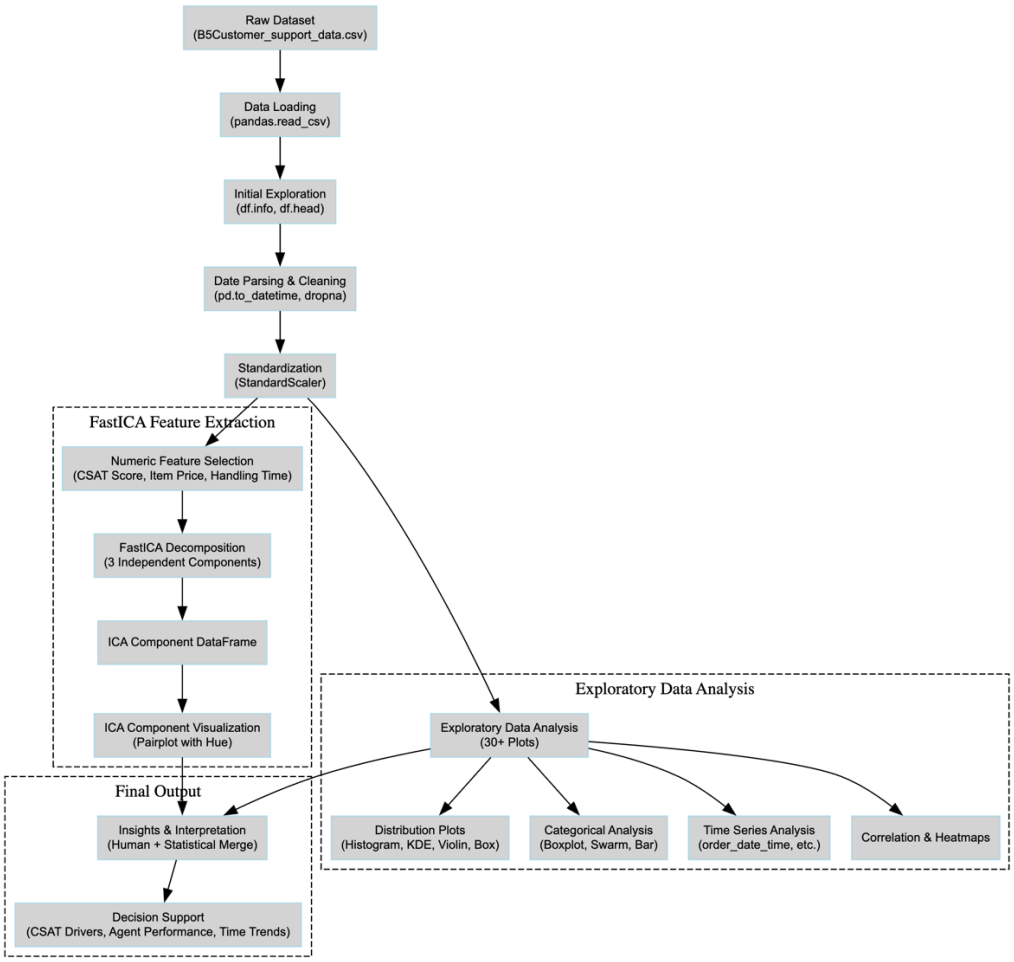

Fig. 4.1: Block diagram of proposed system

Step 1: Data Loading and Preprocessing: Load the

customer support dataset and explore the structure using df.info (). Convert necessary date fields into datetime format and remove or impute missing values to ensure consistency across the dataset. Standardize numerical columns for later dimensionality reduction stages.

Step 2: Exploratory Data Analysis (EDA): Conduct an extensive visual inspection using 30 different plots to uncover trends, anomalies, and distributions. These include histograms, KDE plots, scatter plots, violin plots, swarmplots, boxplots, ECDF, and ridgeline plots. Such comprehensive visual mapping helps understand how satisfaction varies across categories, pricing, handling time, agents, and timeframes.

Step 3: Correlation and Time Series Analysis: Generate correlation heatmaps and time series plots for variables such as order date and response date to observe how CSAT scores evolve over time and correlate with operational metrics. This step aims to capture temporal and operational influencers on satisfaction metrics.

Step 4: Dimensionality Reduction Using FastICA: Apply FastICA on standardized numeric variables, selecting three key features that influence satisfaction: CSAT Score, Item Price, and Connected Handling Time. FastICA extracts hidden, statistically independent components, which can reveal non-linear relationships beyond what standard correlation detects.

Step 5: Visualization of ICA Components: Plot the resulting independent components in a pairwise manner with CSAT score as hue. This enables observation of how hidden factors relate to customer satisfaction clusters and helps in segmenting satisfaction drivers across a high-dimensional operational space.

Step 6: Insight Integration and Recommendation Generation: Interpret the ICA-derived features alongside findings from EDA. Map significant visual and statistical insights to practical logistics and customer support strategies. Formulate improvement areas and decision support systems tailored to the patterns discovered. This hybrid methodology is effective for understanding multifactorial satisfaction influences in customer support settings where both human behaviour and service operations are complex and data-rich.

4.2 Preprocessing

Data preprocessing is a foundational step in Exploratory Data Analysis (EDA) that involves preparing raw data for meaningful insights and visual exploration. The goal is to clean, organize, and transform the data into a structured format that reveals underlying patterns, relationships, and anomalies. This process includes handling missing values, correcting data types, encoding categorical variables, transforming features, and integrating multiple sources. Effective preprocessing enhances the quality of visualizations, simplifies analysis, and ensures more accurate interpretations during the early stages of data exploration.

Step-1: Data Cleaning

- Detect and handle missing values using methods like removal, mean/median imputation, or placeholder values depending on context.
- Identify and remove duplicate records to ensure data consistency.
- Handle outliers using visualization tools like box plots or scatter plots, or statistical methods such as z-scores and interquartile range (IQR).

Step-2: Data Type Conversion

- Convert data types appropriately, e.g., changing date strings to datetime objects or numeric-looking text to actual numbers.
- Ensures correct operations and plots, especially with time series or categorical data.

Step-3: Handling Categorical Data

- Identify and label categorical variables for grouping, filtering, and visualization.
- Convert categorical columns to category type in Pandas to optimize memory and enable grouped analysis.

Step-4: Feature Transformation

- Apply transformations such as log, square root, or power transforms to reduce skewness or make patterns more visible.
- Useful for improving the visibility of distributions and relationships in plots.

Step-5: Encoding Date and Time Features

- Extract features like year, month, day, weekday, or hour from datetime fields.
- Enables time-based grouping, trend detection, and seasonal pattern analysis in visualizations.

Step-6: Data Integration

- Merge or join multiple related datasets using common keys to enrich the data.
- Useful for combining features like customer info with transaction records to enable multi-dimensional analysis.

Step-7: Scaling

- Normalize or standardize features temporarily for better visual comparison, especially when plotting variables on the same scale.
- Though not always required for EDA, scaling helps in visualizing patterns in high variance data.

Step-8: Initial Feature Reduction

- Drop irrelevant, constant, or highly correlated features to simplify plots and avoid visual noise.
- Helps focus EDA on the most informative variables.

These preprocessing steps help ensure that your data is clean, structured, and ready for meaningful exploration using visual and statistical methods.

4.3 EDA

Step-1: Histogram of CSAT Score: The histogram provides a frequency distribution of the CSAT Score, showing how the scores are spread across different bins. The inclusion of the KDE (Kernel Density Estimation) helps understand the smooth distribution of scores, offering insights into skewness or peaks. This plot is essential for identifying the most common satisfaction scores and any extreme values.

Step-2: Boxplot of CSAT Score: The boxplot visualizes the central tendency, dispersion, and outliers in the CSAT Score. The box represents the interquartile range (IQR), with the median marked, while the whiskers show the score range. Outliers are identified as dots outside the whiskers, signaling customers with extremely high or low satisfaction.

Step-3: Violin Plot of CSAT Score: Combining a boxplot with KDE, the violin plot illustrates both the distribution and probability density of CSAT Scores. It reveals multimodal distributions (multiple peaks) and offers a richer view than standard boxplots when dealing with skewed data.

Step-4: KDE Plot of CSAT Score: This standalone KDE plot offers a smoothed curve of the score distribution. It's particularly useful in understanding the underlying probability density function and reveals fine-grained variations in the CSAT Score, such as multiple peaks or a long tail.

Step-5: Bar Plot of CSAT Score Distribution: The bar plot displays the count of each CSAT Score value, highlighting how often each rating appears. This helps identify whether most customers gave high or low ratings and is crucial for evaluating overall service quality.

Step-6: Scatter Plot: Item_price vs CSAT Score: This scatter plot visualizes the relationship between the price of an item and the satisfaction score. It helps detect potential trends or clusters, such as whether higher prices lead to lower satisfaction or if there's no correlation.

Step-7: Scatter Plot: connected_handling_time vs CSAT Score: This plot helps assess if longer or shorter handling times influence CSAT Scores. It might show whether efficiency in handling customer issues correlates positively with customer satisfaction.

Step-8–15: Boxplots: CSAT Score with Categorical Variables: These boxplots analyze how CSAT Score varies across different categorical attributes like category, sub-category, agent shift, tenure bucket, manager, and supervisor. Each plot shows the spread and median CSAT Score per category, highlighting which groups perform better or worse in customer satisfaction.

Step-16: Bar Plot: CSAT Score by Channel Name: This plot shows the average CSAT Score per customer communication channel (e.g., phone, email). It helps identify which channels yield higher customer satisfaction, guiding process improvements.

Step-17–19: Time Series Analysis: These line plots track the average CSAT Score over time for key timestamps (order date, survey response date, issue reported at). They reveal seasonal trends, anomalies, or shifts in satisfaction levels over time, supporting operational adjustments.

Step-20: Heatmap of CSAT Score Correlation: This heatmap shows the correlation matrix between CSAT Score, item price, and handling time. It visually presents the strength and direction of linear relationships, helping identify which numerical factors are most associated with customer satisfaction.

Step-21: Pairplot of Numerical Variables: The pairplot displays scatterplots and distributions for each pair of numerical variables, offering an in-depth look into linear and non-linear relationships. It's useful for spotting patterns or clusters among CSAT Score, item price, and handling time.

Step-22: Swarmplot of CSAT Score by Category: The swarmplot displays individual data points categorized by item category, showing how satisfaction scores vary across product types. It's useful for analyzing specific product performance in relation to customer sentiment.

Step-27: Pie Chart of CSAT Score Distribution: This pie chart presents the percentage share of each CSAT Score. It provides an intuitive and immediate understanding of customer sentiment distribution, though it may be less effective than bar plots for detailed comparison.

Step-28: Boxen Plot of CSAT Score by Agent Name: The boxen plot emphasizes the distribution of CSAT Scores for each agent, focusing on the tails of the distribution. It highlights agents with high consistency or those causing dissatisfaction, which is useful for performance evaluation.

Step-29: ECDF Plot of CSAT Score: The Empirical Cumulative Distribution Function plot shows the proportion of data points below each CSAT Score. It offers a cumulative view of distribution and is ideal for percentile-based analysis.

Step-30: Ridgeline Plot: CSAT Score per Category: This KDE-based ridgeline plot overlays multiple distributions (one per category) to show how customer satisfaction differs across item types. It is great for comparative analysis and identifying which categories have consistent or skewed satisfaction ratings.

4.4 FastICA

Step 1: Numerical Feature Selection: First, we begin by identifying the relevant numerical features from the dataset. For this analysis, these include CSAT Score, Item_price, and connected_handling_time. These features are assumed to carry useful information for detecting hidden patterns in customer satisfaction.

Step 2: Removing Missing Data: Before performing any transformations, we remove all rows with missing values in the selected numerical columns. This is necessary because FastICA requires complete data for proper computation.

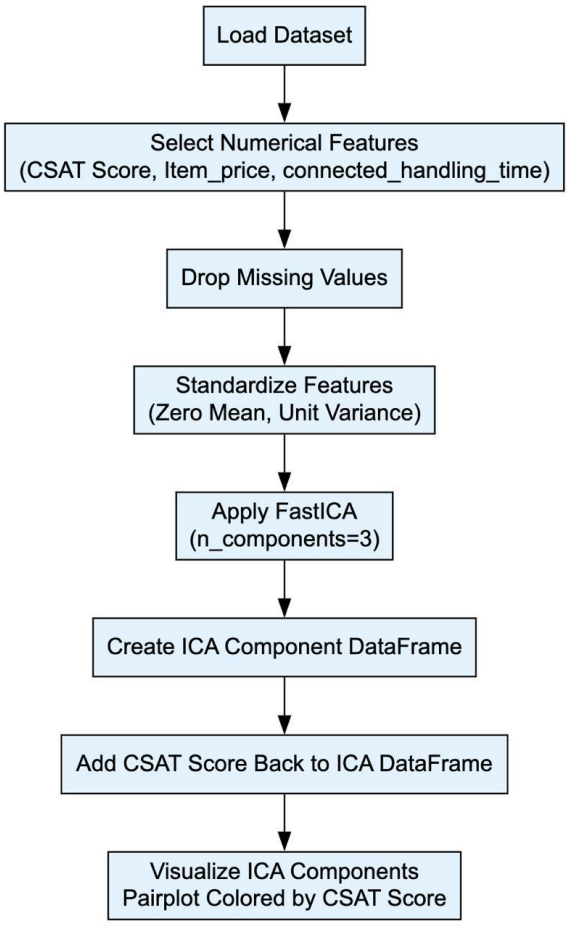

Fig. 4.4: Block diagram of FastICA.

Step 3: Standardization of Features: The selected numerical data is then standardized—meaning each feature is adjusted to have a mean of zero and standard deviation of one. This ensures that all variables contribute equally to the FastICA process and that scale differences don't skew the results.

Step 4: Applying FastICA: Now we apply Fast Independent Component Analysis (FastICA)**.** This technique tries to decompose the dataset into independent, non-Gaussian components. These components represent hidden features or sources that are statistically independent from each other.

Step 5: Creating a Transformed Dataset: decomposition, the resulting components are organized into a new dataset. Each row now represents a customer interaction in terms of these

independent signals (ICA1, ICA2, ICA3)**.** We also retain the original CSAT Score for comparison and visualization.

Step 6: Visualization: Finally, we use visualizations like pairplots to explore how these ICA components interact with each other and with the CSAT Score. Color gradients help highlight if certain independent features are strongly associated with high or low satisfaction levels.

4.5 Feature Importance Plots

Pairplot of ICA Components with CSAT Score Hue: This plot visualizes the independent components extracted using FastICA and overlays them with color-coded CSAT Scores. By observing how different CSAT Score clusters appear in the ICA space, one can intuitively understand which combinations of the original variables (item price, handling time) are most influential. The spread and overlap of points help identify whether the independent features effectively separate customer sentiment levels, highlighting potential for clustering or predictive modeling.

CHAPTER 5

UML DIAGRAMS

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group. The goal is for UML to become a common language for creating models of object-oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems. The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems. The UML is a very important part of developing objects-oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

GOALS: The Primary goals in the design of the UML are as follows:

- Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
- Provide extendibility and specialization mechanisms to extend the core concepts.
- Be independent of particular programming languages and development process.
- Provide a formal basis for understanding the modeling language.
- Encourage the growth of OO tools market.
- Support higher level development concepts such as collaborations, frameworks, patterns and components.
- Integrate best practices.

5.1 Class diagram

The class diagram is used to refine the use case diagram and define a detailed design of the system. The class diagram classifies the actors defined in the use case diagram into a set of interrelated classes. The relationship or association between the classes can be either an "is-a"

or "has-a" relationship. Each class in the class diagram was capable of providing certain functionalities. These functionalities provided by the class are termed "methods" of the class. Apart from this, each class may have certain "attributes" that uniquely identify the class.

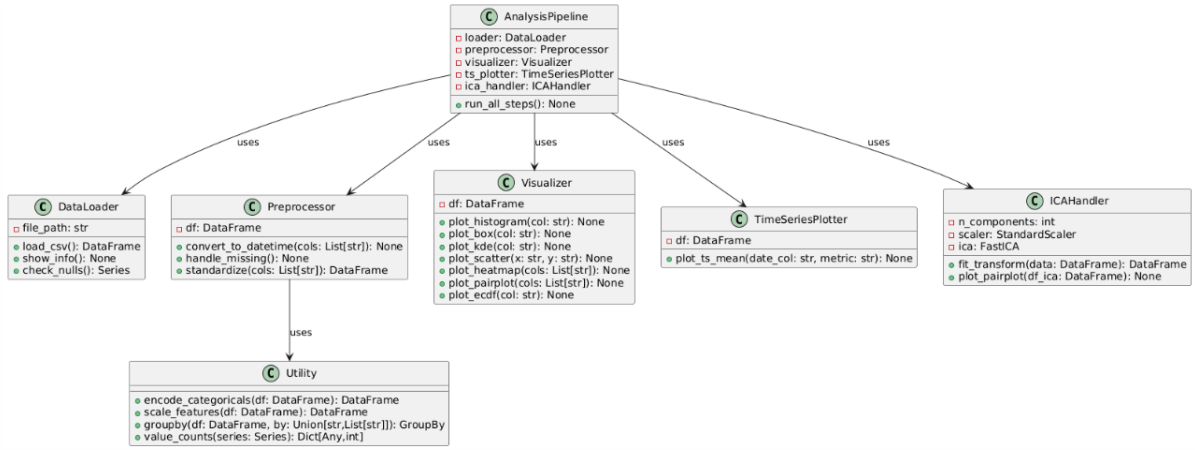

Fig. 5.1: Class Diagram

5.2 Sequence Diagram

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. A sequence diagram shows, as parallel vertical lines (“lifelines”), different processes or objects that live simultaneously, and as horizontal arrows, the messages exchanged between them, in the order in which they occur. This allows the specification of simple runtime scenarios in a graphical manner.

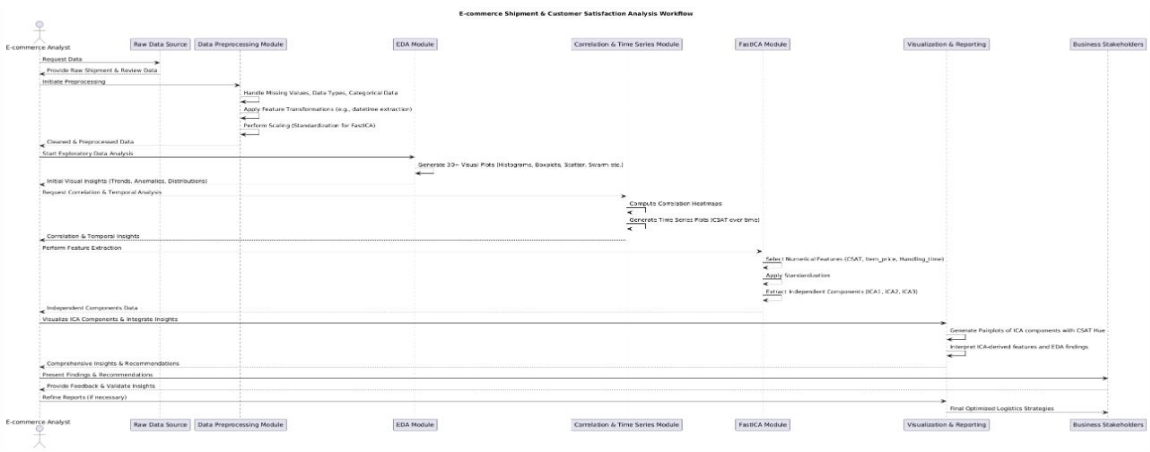

Fig. 5.2: Sequence Diagram

5.3 Data flow diagram

A data flow diagram (DFD) is a graphical representation of how data moves within an information system. It is a modeling technique used in system analysis and design to illustrate the flow of data between various processes, data stores, data sources, and data destinations within a system or between systems. Data flow diagrams are often used to depict the structure and behavior of a system, emphasizing the flow of data and the transformations it undergoes as it moves through the system.

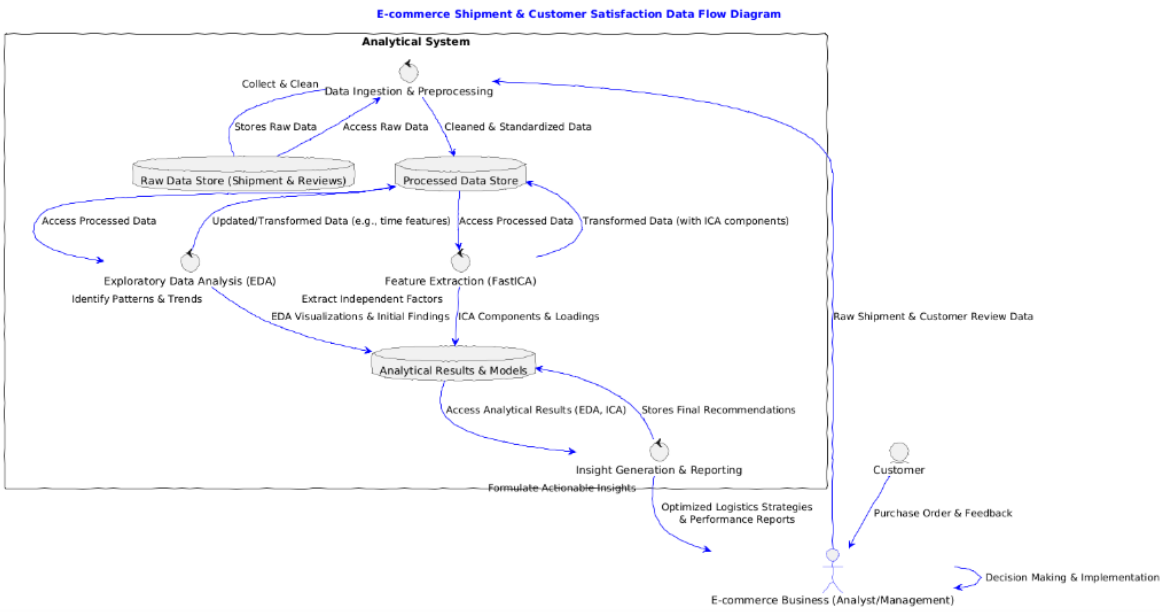

Fig. 5.3: Data Flow Diagram

5.4 Activity diagram

Activity diagrams are graphical representations of Workflows of stepwise activities and actions with support for choice, iteration, and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.

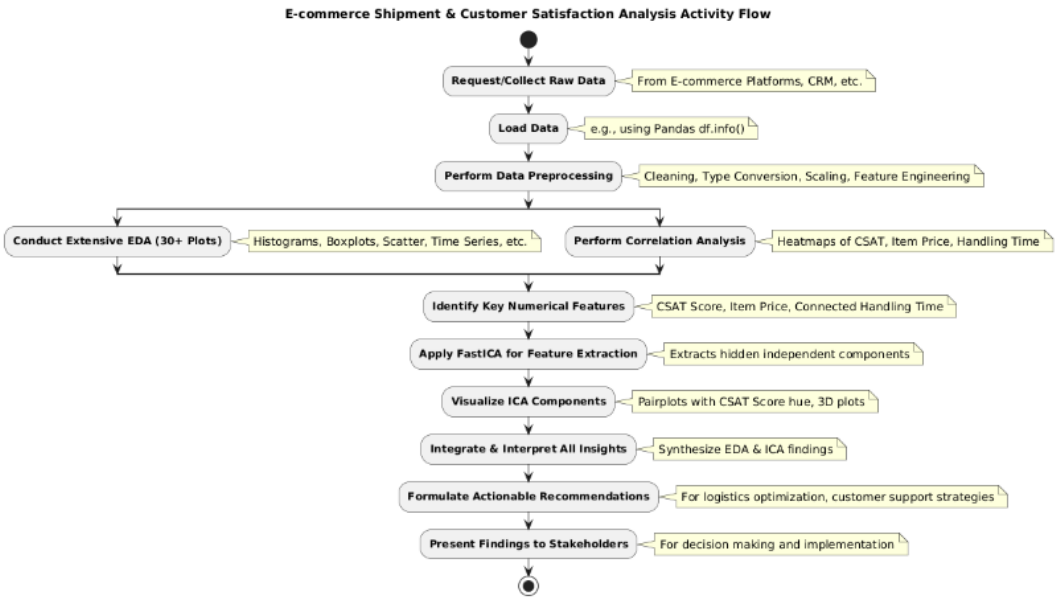

Fig. 5.4: Activity Diagram

5.5 Component diagram: Component diagram describes the organization and wiring of the physical components in a system.

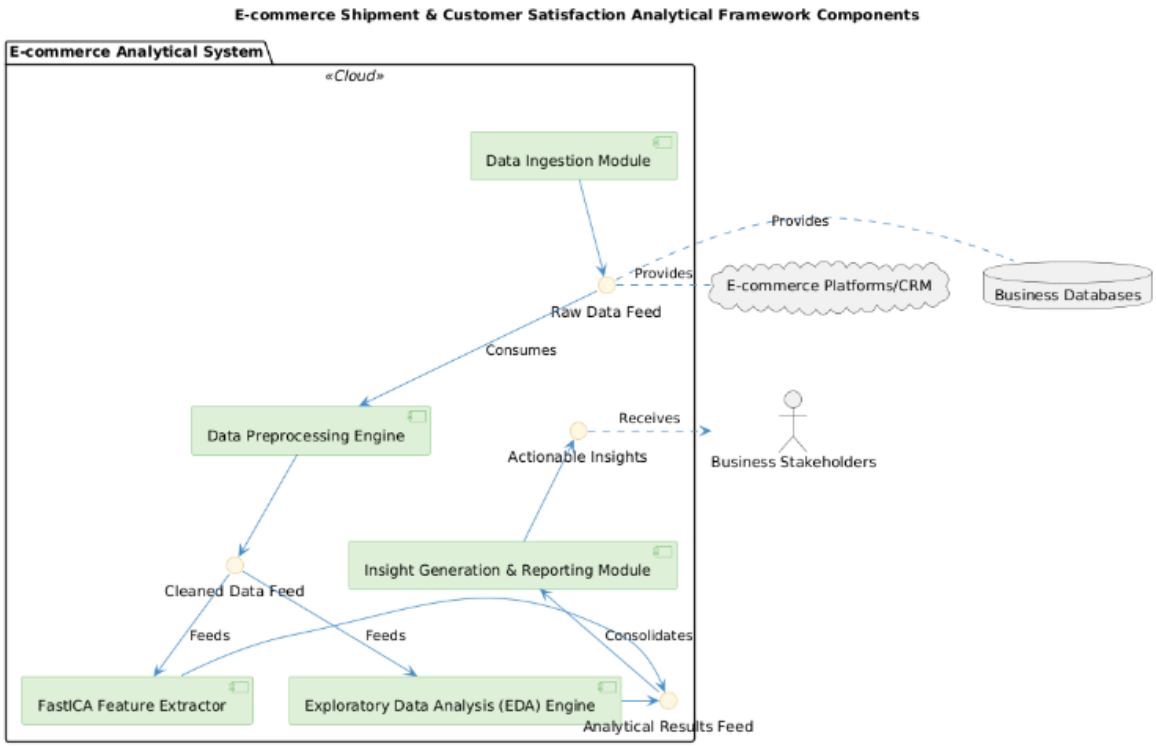

Fig. 5.5: Component Diagram

5.6 Use Case diagram:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

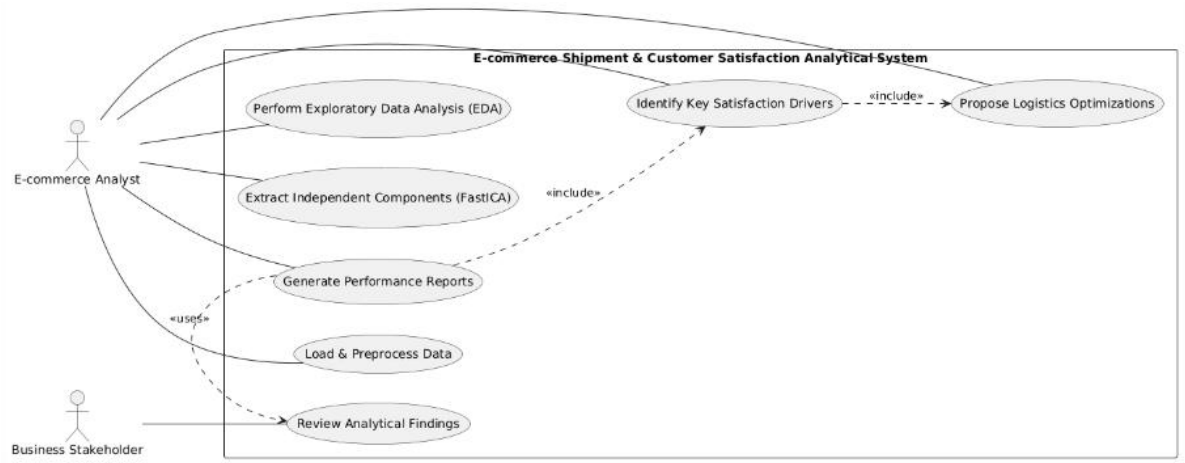

Fig. 5.6: Use Case Diagram

CHAPTER 6
SOFTWARE DESCRIPTION

6.1 Software Requirements

Python is a high-level, interpreted programming language known for its simplicity and readability, which makes it a popular choice for beginners as well as experienced developers. Key features of Python include its dynamic typing, automatic memory management, and a rich standard library that supports a wide range of applications from web development to data science and machine learning. Its object-oriented approach and support for multiple programming paradigms allow developers to write clear, maintainable code. Python's extensive ecosystem of third-party packages further enhances its capabilities, enabling rapid development and prototyping across diverse fields.

Installation

First, download the appropriate installer from the official Python website (https://www.python.org/downloads/release/python-376/). For Windows users, run the executable installer and ensure to check the "Add Python to PATH" option during installation; for macOS and Linux, follow the respective package installation commands or use a package manager like Homebrew or apt-get. After installation, verify the setup by running python --version or python3 --version in your terminal or command prompt, which should display "Python 3.7.6." This version-specific installation supports all major functionalities and libraries compatible with Python 3.7.6, making it an excellent foundation for developing robust applications in areas such as data analysis, machine learning, and GUI development.

6.1.1 Python Packages

The project requires a robust set of software libraries and tools. Below is an explanation of the key software requirements and the packages used:

- **Python:** The project is implemented in Python, which is chosen for its extensive ecosystem of libraries and its strong support for data analysis, machine learning, and GUI development.

- **Matplotlib:**

➢ Core Visualization Library:

  - Serves as the foundational Python library for creating static, animated, and interactive visualizations.
  - Widely used for plotting line charts, scatter plots, bar charts, histograms, and more.

➢ Customization & Flexibility:

  - Offers granular control over every element of a figure, from plot size and axis labels to colors and styles.
  - Facilitates the creation of publication-quality graphics, making it ideal for fine-tuning visuals during EDA.

➢ Integration:

  - Easily integrates with other libraries like NumPy and Pandas for quick plotting of data arrays and dataframes.
  - Acts as the underlying engine for other visualization libraries (e.g., Seaborn), extending its functionality.

- **Seaborn:**

➢ Statistical Data Visualization:

  - Built on Matplotlib, Seaborn specializes in making statistical plots that are informative and aesthetically appealing.
  - Provides high-level interfaces for drawing attractive and complex visualizations such as violin plots, box plots, heatmaps, and pair plots.

➢ Simplified Syntax for Complex Graphics:

  - Enables quick and easy generation of plots with simple function calls, abstracting many of the lower-level details.

- Automatically handles complex data structures and applies effective defaults, reducing the manual adjustments needed during EDA.

➢ EDA-Focused Enhancements:

- Designed to seamlessly work with Pandas data structures for exploring datasets.
- Enhances the ability to quickly spot patterns, outliers, and relationships in the data through visually engaging plots.

- **Plotly**

➢ Interactive and Web-Based Visualizations:

- Offers a modern approach to EDA by providing highly interactive plots that allow users to zoom, pan, and hover over data points to inspect details.
- Ideal for creating dashboards and sharing insights with a broader audience using interactive web applications.

➢ Versatility and Aesthetics:

- Supports a wide variety of plot types including line plots, scatter plots, bar charts, 3D plots, and even geographical maps.
- Enhances the exploratory process by providing immediate visual feedback and dynamic interactions with the data.

➢ Ease of Integration with Modern Web Frameworks:

- Works well in Jupyter notebooks as well as in web applications, allowing seamless integration into data science workflows.
- Can be used in conjunction with other Python libraries to enhance EDA by adding interactive layers to static plots.

- **Scikit-learn**

➢ Comprehensive Machine Learning Toolkit:

- Provides a consistent and well-documented API for various machine learning tasks, including unsupervised learning.

- Offers a wide range of unsupervised learning algorithms such as clustering (K-Means, DBSCAN, Agglomerative Clustering) and dimensionality reduction (Principal Component Analysis, t-SNE).

- Integration with EDA Workflows:
  - Allows you to preprocess and transform datasets before visualizing them, facilitating robust EDA.
  - Unsupervised learning techniques help reveal hidden structures in the data, like groups or clusters, which can be further explored using visualizations from Matplotlib, Seaborn, or Plotly.

- Pipeline and Model Evaluation:
  - Encourages the use of pipelines to chain together preprocessing, dimensionality reduction, and clustering, streamlining the EDA process.
  - Provides tools for model evaluation and cross-validation, ensuring that the exploratory insights gained from unsupervised learning are statistically significant and reproducible.

These packages collectively cover a broad spectrum of tasks in the EDA process. Matplotlib and Seaborn are excellent for static and quick visualizations, Plotly adds an interactive dimension to data exploration, and Scikit-learn offers advanced unsupervised learning capabilities to uncover underlying structures in your data. Together, they empower data scientists to perform thorough and insightful exploratory data analyses.

6.2 Hardware Requirements

Python 3.7.6 can run efficiently on most modern systems with minimal hardware requirements. However, meeting the recommended specifications ensures better performance, especially for developers handling large-scale applications or computationally intensive tasks. By ensuring compatibility with hardware and operating system, can leverage the full potential of Python 3.7.6.

Processor (CPU) Requirements: Python 3.7.6 is a lightweight programming language that can run on various processors, making it highly versatile. However, for optimal performance, the following processor specifications are recommended:

- **Minimum Requirement**: 1 GHz single-core processor.
- **Recommended**: Dual-core or quad-core processors with a clock speed of 2 GHz or higher. Using a multi-core processor allows Python applications, particularly those involving multithreading or multiprocessing, to execute more efficiently.

Memory (RAM) Requirements: Python 3.7.6 does not demand excessive memory but requires adequate RAM for smooth performance, particularly for running resource-intensive applications such as data processing, machine learning, or web development.

- **Minimum Requirement**: 512 MB of RAM.
- **Recommended**: 4 GB or higher for general usage. For data-intensive operations, 8 GB or more is advisable.

Insufficient RAM can cause delays or crashes when handling large datasets or executing computationally heavy programs.

Storage Requirements: Python 3.7.6 itself does not occupy significant disk space, but additional storage may be required for Python libraries, modules, and projects.

- **Minimum Requirement**: 200 MB of free disk space for installation.
- **Recommended**: At least 1 GB of free disk space to accommodate libraries and dependencies.

Developers using Python for large-scale projects or data science should allocate more storage to manage virtual environments, datasets, and frameworks like TensorFlow or PyTorch.

Compatibility with Operating Systems: Python 3.7.6 is compatible with most operating systems but requires hardware that supports the respective OS. Below are general requirements for supported operating systems:

- **Windows**: 32-bit and 64-bit systems, Windows 7 or later.
- **macOS**: macOS 10.9 or later.
- **Linux**: Supports a wide range of distributions, including Ubuntu, CentOS, and Fedora.

The hardware specifications for the OS directly impact Python's performance, particularly for modern software development.

CHAPTER 7
FUNCTIONAL REQUIREMENTS

7.1. Introduction

This document outlines the functional requirements for the "E-commerce Shipment and Customer Satisfaction Analytical System." These requirements specify the capabilities and behaviors that the system must exhibit to achieve its objectives of providing actionable insights from shipment and customer feedback data to enhance overall customer satisfaction.

7.2 Functional Requirements

The E-commerce Shipment and Customer Satisfaction Analytical System shall provide the following functionalities:

FR1: Data Ingestion and Management This set of requirements details the system's capabilities for acquiring, cleaning, transforming, and preparing raw data for analysis.

- **FR1.1 Data Loading:** The system shall enable the analyst to load raw customer support datasets, encompassing shipment details, customer reviews, Customer Satisfaction (CSAT) scores, item prices, and handling times from specified sources.
- **FR1.2 Data Structure Exploration:** The system shall provide a mechanism to display the structural overview and basic metadata of the loaded dataset (e.g., column names, data types, non-null counts).
- **FR1.3 Date Field Conversion:** The system shall automatically convert relevant date and time fields (e.g., order date, survey response date, issue reported at) into a standard datetime format suitable for temporal analysis.
- **FR1.4 Missing Value Handling:** The system shall provide configurable methods (e.g., removal of rows/columns, mean/median/mode imputation, placeholder values) for detecting and managing missing values within the dataset.
- **FR1.5 Duplicate Record Removal:** The system shall identify and remove duplicate records to ensure the uniqueness and consistency of the dataset.
- **FR1.6 Outlier Handling:** The system shall offer tools for detecting and managing outliers in numerical features, including visualization-based identification (e.g., box plots, scatter plots) and statistical methods (e.g., Z-scores, IQR).

- **FR1.7 Data Type Conversion:** The system shall ensure correct data type assignment for all columns (e.g., converting numeric-looking text to actual numerical types, and categorical strings to optimized category types in Pandas).
- **FR1.8 Feature Transformation:** The system shall support applying mathematical transformations (e.g., logarithmic, square root, power transforms) to numerical features to reduce skewness or improve interpretability of relationships.
- **FR1.9 Date and Time Feature Encoding:** The system shall extract and encode granular temporal features (e.g., year, month, day of week, hour of day) from datetime fields to facilitate time-based grouping and trend analysis.
- **FR1.10 Data Integration:** The system shall support the merging or joining of multiple related datasets using common keys to enrich the analytical dataset.
- **FR1.11 Feature Scaling:** The system shall standardize selected numerical features to have a mean of zero and a standard deviation of one, which is a prerequisite for algorithms like FastICA.
- **FR1.12 Initial Feature Reduction:** The system shall allow the analyst to drop irrelevant, constant, or highly correlated features to streamline analysis and improve visualization clarity.

FR2: Exploratory Data Analysis (EDA) This set of requirements specifies the system's capabilities for comprehensive data visualization and initial pattern discovery.

- **FR2.1 CSAT Score Distribution Visualization:** The system shall generate a variety of visual plots (e.g., histograms with KDE, boxplots, violin plots, standalone KDE plots, bar plots, pie charts, ECDF plots) to illustrate the frequency distribution, central tendency, dispersion, and cumulative density of CSAT scores.
- **FR2.2 Numerical Variable Relationship Visualization:** The system shall generate scatter plots to visualize the relationships between CSAT Score and other key numerical variables such as Item_price and connected_handling_time.
- **FR2.3 Categorical Variable Impact Visualization:** The system shall generate boxplots and swarmplots to visualize how CSAT Score distributions vary across different categorical attributes (e.g., product category, sub-category, agent shift, agent tenure bucket, manager, supervisor, customer communication channel name).

- **FR2.4 Temporal Trend Analysis:** The system shall generate time series plots to track the average CSAT Score over time based on critical timestamps (e.g., order date, survey response date, issue reported at) to identify seasonal patterns or shifts.
- **FR2.5 Correlation Visualization:** The system shall generate correlation heatmaps to visually represent the strength and direction of linear relationships between CSAT Score, Item_price, and connected_handling_time.
- **FR2.6 Pairwise Numerical Relationship Visualization:** The system shall generate pairplots to display scatterplots and marginal distributions for each pair of relevant numerical variables, providing a holistic view of their relationships.
- **FR2.7 Comparative Distribution Visualization:** The system shall generate ridgeline plots to effectively compare the CSAT Score distributions across different categorical item types.
- **FR2.8 Agent Performance Visualization:** The system shall generate boxen plots to provide detailed insights into the distribution of CSAT Scores for individual agents, highlighting consistency and areas of concern.

FR3: Feature Extraction and Interpretation This set of requirements focuses on the system's advanced analytical capabilities to uncover latent factors.

- **FR3.1 Numerical Feature Selection for FastICA:** The system shall allow the analyst to specify the set of numerical features (e.g., CSAT Score, Item_price, connected_handling_time) to be used as input for the FastICA algorithm.
- **FR3.2 FastICA Application:** The system shall apply the Fast Independent Component Analysis (FastICA) algorithm on the selected and standardized numerical features to extract a specified number of hidden, statistically independent components.
- **FR3.3 ICA Component Visualization:** The system shall generate visual plots (e.g., pairplots with CSAT score as hue, 3D component visualizations) to illustrate the relationships between the extracted ICA components and their association with customer satisfaction levels.
- **FR3.4 Insight Integration:** The system shall provide a framework for integrating and interpreting the combined insights derived from both the comprehensive EDA and the FastICA-extracted features.

FR4: Reporting and Recommendation Generation This set of requirements defines how the analytical findings are translated into actionable intelligence for business stakeholders.

- **FR4.1 Recommendation Formulation Support:** The system shall support the analyst in formulating practical and data-backed logistics and customer support strategies based on the identified patterns and insights from the analysis.
- **FR4.2 Decision Support Output:** The system shall generate outputs (e.g., summarized reports, interactive dashboards) that directly serve as a decision support system for e-commerce businesses to address critical customer satisfaction issues.
- **FR4.3 Comprehensive Analytical Reporting:** The system shall enable the generation of comprehensive reports detailing the analytical methodology, findings, key visualizations, and derived recommendations for business stakeholders.

CHAPTER 8

SOURCE CODE

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import plotly.graph_objects as go
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler

path = r"A2train_schedule.csv"
df = pd.read_csv(path)

# Columns of interest
cols = ["1A", "2A", "3A", "SL", "Distance"]

# Compute Min, Max, Mean for each column
for col in cols:
    df[f"{col}_Min"] = df[col].min()
    df[f"{col}_Max"] = df[col].max()
    df[f"{col}_Mean"] = df[col].mean()

df.info()

# Plotting histograms and KDE
fig, axes = plt.subplots(3, 2, figsize=(15, 20))
columns = ["Distance", "1A", "2A", "3A", "SL"]
for i, col in enumerate(columns):
    sns.histplot(df[col], bins=30, kde=True, ax=axes[i//2, i%2])
    axes[i//2, i%2].set_title(f"Distribution of {col}")
plt.tight_layout()
plt.show()
```

```
# Plotting boxplots
fig, axes = plt.subplots(1, 5, figsize=(20, 5))
columns = ["Distance", "1A", "2A", "3A", "SL"]
for i, col in enumerate(columns):
    sns.boxplot(y=df[col], ax=axes[i])
    axes[i].set_title(f"Boxplot of {col}")
plt.show()

# Arrival and Departure Hour
df["Arrival_Hour"] = pd.to_datetime(df["Arrival_time"], errors="coerce").dt.hour
df["Departure_Hour"] = pd.to_datetime(df["Departure_Time"], errors="coerce").dt.hour
fig, axes = plt.subplots(2, 2, figsize=(15, 10))
sns.histplot(df["Arrival_Hour"].dropna(), bins=24, kde=True, ax=axes[0, 0])
axes[0, 0].set_title("Arrival Time Distribution")
sns.histplot(df["Departure_Hour"].dropna(), bins=24, kde=True, ax=axes[0, 1])
axes[0, 1].set_title("Departure Time Distribution")
sns.boxplot(x=df["Arrival_Hour"], y=df["1A_Mean"], ax=axes[1, 0])
axes[1, 0].set_title("1A Mean Price vs Arrival Hour")
sns.boxplot(x=df["Departure_Hour"], y=df["1A_Mean"], ax=axes[1, 1])
axes[1, 1].set_title("1A Mean Price vs Departure Hour")
plt.show()

# Pie Charts
fig, axes = plt.subplots(1, 3, figsize=(18, 5))
df[["1A_Mean", "2A_Mean", "3A_Mean"]].mean().plot.pie(autopct="%1.1f%%",
ax=axes[0], title="Fare Class Distribution (Mean)")
df[["SL_Mean", "Distance_Mean"]].mean().plot.pie(autopct="%1.1f%%", ax=axes[1],
title="Sleeper & Distance Price Distribution")
df["Route_Number"].value_counts().nlargest(5).plot.pie(autopct="%1.1f%%", ax=axes[2],
title="Top 5 Routes by Frequency")
plt.show()

# Ticket Price Range Boxplots
```

```
fig5 = go.Figure()
fig5.add_trace(go.Box(y=df["1A_Min"], name="1A Min"))
fig5.add_trace(go.Box(y=df["1A_Max"], name="1A Max"))
fig5.add_trace(go.Box(y=df["1A_Mean"], name="1A Mean"))
fig5.update_layout(title="1A Ticket Price Range")
fig5.show()

fig6 = go.Figure()
fig6.add_trace(go.Box(y=df["2A_Min"], name="2A Min"))
fig6.add_trace(go.Box(y=df["2A_Max"], name="2A Max"))
fig6.add_trace(go.Box(y=df["2A_Mean"], name="2A Mean"))
fig6.update_layout(title="2A Ticket Price Range")
fig6.show()

fig7 = go.Figure()
fig7.add_trace(go.Box(y=df["3A_Min"], name="3A Min"))
fig7.add_trace(go.Box(y=df["3A_Max"], name="3A Max"))
fig7.add_trace(go.Box(y=df["3A_Mean"], name="3A Mean"))
fig7.update_layout(title="3A Ticket Price Range")
fig7.show()

fig8 = go.Figure()
fig8.add_trace(go.Box(y=df["SL_Min"], name="SL Min"))
fig8.add_trace(go.Box(y=df["SL_Max"], name="SL Max"))
fig8.add_trace(go.Box(y=df["SL_Mean"], name="SL Mean"))
fig8.update_layout(title="SL Ticket Price Range")
fig8.show()

# Distance vs. Ticket Price (1A)
fig9 = px.scatter(df, x="Distance", y="1A", title="Distance vs. 1A Ticket Price")
fig9.show()

# PCA and KMeans Clustering
```

```
num_cols = ["Distance", "1A_Mean", "2A_Mean", "3A_Mean", "SL_Mean", "Distance_Min",
"Distance_Max", "Distance_Mean"]
df_num = df[num_cols]
scaler = StandardScaler()
df_scaled = scaler.fit_transform(df_num)
pca = PCA(n_components=3)
pca_data = pca.fit_transform(df_scaled)
df_pca = pd.DataFrame(pca_data, columns=["PC1", "PC2", "PC3"])
kmeans = KMeans(n_clusters=4, random_state=42, n_init=10)
df_pca["Cluster"] = kmeans.fit_predict(df_pca)

df_combined = df.join(df_pca)

# PCA Explained Variance Ratio
explained_variance = pca.explained_variance_ratio_ * 100
fig1 = px.bar(x=["PC1", "PC2", "PC3"], y=explained_variance, title="PCA Explained
Variance Ratio", labels={'y': "Variance (%)", 'x': "Principal Components"})
fig1.show()

# PCA Clustering (PC1 vs PC2)
fig2 = px.scatter(df_combined, x="PC1", y="PC2", color=df_combined["Cluster"].astype(str),
title="PCA Clustering (PC1 vs PC2)")
fig2.show()

# 3D PCA Clustering
fig3 = px.scatter_3d(df_combined, x="PC1", y="PC2", z="PC3",
color=df_combined["Cluster"].astype(str), title="3D PCA Clustering")
fig3.show()

# Cluster Count Distribution
fig4 = px.histogram(df_combined, x="Cluster", title="Cluster Distribution", nbins=10)
fig4.show()

# Distance vs. PC1 (with Clusters)
```

```
fig5 = px.scatter(df_combined, x="Distance", y="PC1", color=df_combined["Cluster"].astype(str), title="Distance vs. PC1 (Clustered)")
fig5.show()
```

CHAPTER 9

RESULTS AND DISCUSSION

9.1 Implementation Description

Below is a step-wise explanation of implementation description of our project:

Step-1: Data Loading and Setup

- The dataset B5Customer_support_data.csv is loaded into a Pandas DataFrame named df.
- Basic dataset insights are retrieved using df.info () to understand data types and null counts.

Step-2: Univariate Analysis of CSAT Score

- **Histogram**: Plots the distribution of CSAT Score with KDE (Kernel Density Estimate) to see its frequency and smooth curve.
- **Boxplot**: Highlights outliers and the quartile distribution of CSAT Score.
- **Violin Plot**: Combines boxplot and KDE to visualize the data's spread and density.
- **KDE Plot**: Separately visualizes the continuous density of CSAT Score.
- **Bar Plot (Count Plot)**: Shows the count of each distinct CSAT Score value.

Step-3: Bivariate Relationships with CSAT Score

- **Scatter Plot** between Item_price and CSAT Score: Used to examine linear or non-linear correlations.
- **Scatter Plot** between connected_handling_time and CSAT Score: Assesses service time's impact on satisfaction.

Step-4: Categorical vs CSAT Score Analysis

- Iterates over several categorical features like:
    - category, Sub-category, Agent Shift, Tenure Bucket, Manager, Supervisor
- Uses Boxplots to show how CSAT Scores differ across each category.

Step-5: Channel-based Analysis

- A Bar Plot shows the average CSAT Score for each channel_name (e.g., Email, Phone).

Step-6: Time-Series Analysis

- Converts date columns (order_date_time, Survey_response_Date, Issue_reported at) to datetime format.
- Groups and averages

9.2 Dataset Description

- **Unique id**: A unique identifier for each record or customer interaction.
- **Channel name**: The communication channel used for the customer interaction (e.g., Outcall, Inbound, Email).
- **category**: The broad category of the customer's query or issue (e.g., Product Queries, Order Related, Returns, Cancellation, Feedback, Refund Related, Shopzilla Related, Payments related).
- **Sub-category**: A more specific classification of the customer's issue within the main category (e.g., Life Insurance, Product Specific Information, Installation/demo, Reverse Pickup Enquiry, Not Needed, Fraudulent User, Exchange / Replacement, Missing, General Enquiry, Return request, Delayed, Service Centres Related, Unable to track, Order status enquiry, Invoice request, Priority delivery, Wrong, Seller Cancelled Order, UnProfessional Behaviour, Refund Enquiry, Payment related Queries, Refund Related Issues).
- **Customer Remarks**: Free-text comments or feedback provided by the customer, often indicating satisfaction or dissatisfaction with the service or product.
- **Order_id**: A unique identifier for the customer's order.
- **order_date_time**: The date and time when the order was placed. This field may sometimes be empty.
- **Issue_reported at**: The date and time when the customer's issue was reported.
- **issue_responded**: The date and time when the customer's issue was responded to by the support team.
- **Survey_response_Date**: The date when the customer provided their survey response.
- **Customer_City**: The city of the customer. This field may sometimes be empty.

- **Product_category**: The category of the product involved in the order (e.g., LifeStyle, Electronics, Home Appliances, Mobile, Furniture, Home). This field may sometimes be empty.
- **Item_price**: The price of the item in the order. This field may sometimes be empty.
- **connected_handling_time**: Potentially the time taken to handle the customer's issue or shipment. This field may sometimes be empty.
- **Agent_name**: The name of the customer support agent who handled the interaction.
- **Supervisor**: The name of the supervisor of the agent.
- **Manager**: The name of the manager of the agent.
- **Tenure Bucket**: The tenure group of the agent (e.g., On Job Training, >90, 0-30, 31-60, 61-90).
- **Agent Shift**: The shift during which the agent worked (e.g., Morning, Evening, Split, Afternoon, Night).
- **CSAT Score**: The Customer Satisfaction (CSAT) score given by the customer, typically on a scale of 1 to 5, with 5 being highly satisfied.

9.3 Results Description

Customer Satisfaction and Survey-Related Columns

- CSAT Score: Represents the Customer Satisfaction Score, likely measured on a scale (e.g., 1 to 5). This is the target variable for analysis and is used to gauge how satisfied customers were after support interactions.

	Unique id	channel_name	category	Sub-category	Customer Remarks	Order_id	order_date_time	Issue_reported at	issue_responded	Survey_respon
0	7e9ae164-6a8b-4521-a2d4-58f7c9fff13f	Outcall	Product Queries	Life Insurance	NaN	c27c9bb4-fa36-4140-9f1f-21009254ffdb	NaN	01/08/2023 11:13	01/08/2023 11:47	0
1	b07ec1b0-f376-43b6-86df-ec03da3b2e16	Outcall	Product Queries	Product Specific Information	NaN	d406b0c7-ce17-4654-b9de-f08d421254bd	NaN	01/08/2023 12:52	01/08/2023 12:54	0
2	200814dd-27c7-4149-ba2b-bd3af3092880	Inbound	Order Related	Installation/demo	NaN	c273368d-b961-44cb-beaf-62d6fd6c00d5	NaN	01/08/2023 20:16	01/08/2023 20:38	0
3	eb0d3e53-c1ca-42d3-8486-e42c8d622135	Inbound	Returns	Reverse Pickup Enquiry	NaN	5aed0059-55a4-4ec6-bb54-97942092020a	NaN	01/08/2023 20:56	01/08/2023 21:16	0
4	ba903143-1e54-406c-b969-46c52f92e5df	Inbound	Cancellation	Not Needed	NaN	e8bed5a9-6933-4aff-9dc6-ccefd7dcde59	NaN	01/08/2023 10:30	01/08/2023 10:32	0
...	...	...	...	...	...	...	...	...	...	
85902	505ea5e7-c475-4fac-ac36-1d19a4cb610f	Inbound	Refund Related	Refund Enquiry	NaN	1b5a2b9c-a95f-405f-a42e-5b1b693f3dc9	NaN	30/08/2023 23:20	31/08/2023 07:22	3
85903	44b38d3f-1523-4182-aba2-	Inbound	Order Related	Seller Cancelled Order	Supported team customer	d0e8a817-96d5-4ace-bb82-	NaN	31/08/2023 08:15	31/08/2023 08:17	3

Fig.9.3 a: Sample dataset.

Product and Order-Related Columns

- Item_price:
  The price of the product or item purchased by the customer. This is a numerical feature used to analyze how product cost relates to customer satisfaction.
- category:
  The main product category (e.g., electronics, furniture, etc.) that the purchased item belongs to.
- Sub-category:
  A subdivision of the category, providing more specific grouping of the product (e.g., within electronics: phones, laptops, etc.).

Agent and Employee-Related Columns

- Agent_name:
  Name or identifier of the support agent who handled the customer's issue.
- Agent Shift:

  Indicates the shift timing of the agent (e.g., morning, evening, night), which could influence response quality or availability.

- Manager:
  The manager's name or ID overseeing the support agent.
- **Supervisor**:
  Indicates the supervisor assigned to the support team or interaction.
- Tenure Bucket:

  Represents the experience level or duration of time an agent has been with the company (e.g., "0-1 year", "2-5 years").

Time-Related Columns

- order_date_time:
  The timestamp of when the customer's order was placed. Useful for time series trends.

- Survey_response_Date:
  Date when the customer submitted their CSAT response. Helps track satisfaction across time.
- Issue_reported
  Date and time when the customer reported an issue. Helps align resolution time or urgency patterns.

Operational Metrics

- connected_handling_time:
  Time taken by the support agent to handle the customer's query. This could include total call/chat time and is crucial for performance analysis.

Communication Channel

- channel_name:
  The medium through which the customer contacted support, such as Email, Phone, Live Chat, or SocialMedia. This can affect resolution time and satisfaction.

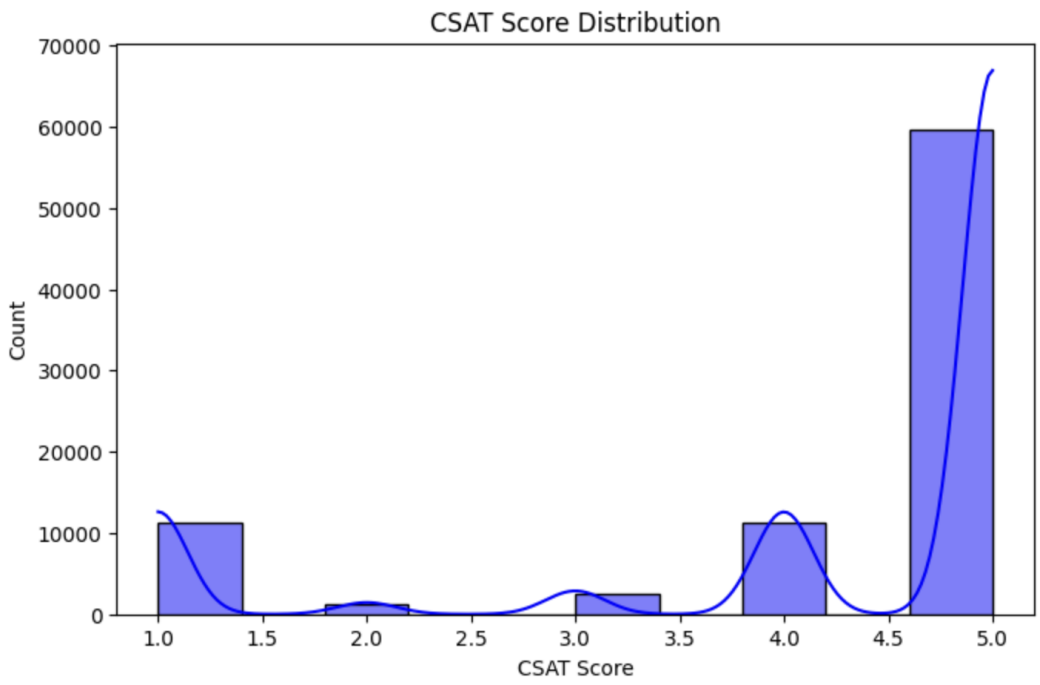

Fig.9.3 b: Histogram of CSAT score distribution.

The Figure 2 displays an "E-commerce Order Fulfilment Process" diagram, illustrating the journey from order placement to final delivery. It begins with the Order Received stage,

followed by Inventory Check & Allocation. If products are in stock, the process moves to Picking & Packing. Next, the order proceeds to Shipment (Carrier) and then Last-Mile Delivery. Finally, it reaches the Customer Received stage, potentially leading to Returns if unsatisfactory. This visual outlines the critical steps in ensuring an order reaches the customer.

The Figure 3 boxplot of CSAT Scores reveals the distribution of customer satisfaction ratings. The central box indicates the interquartile range (IQR), encompassing the middle 50% of scores. The line within the box represents the median CSAT score. Whiskers extend to the minimum and maximum scores within 1.5 times the IQR, while individual points beyond the whiskers denote outliers. This visualization quickly highlights typical satisfaction levels and any extreme customer experiences.

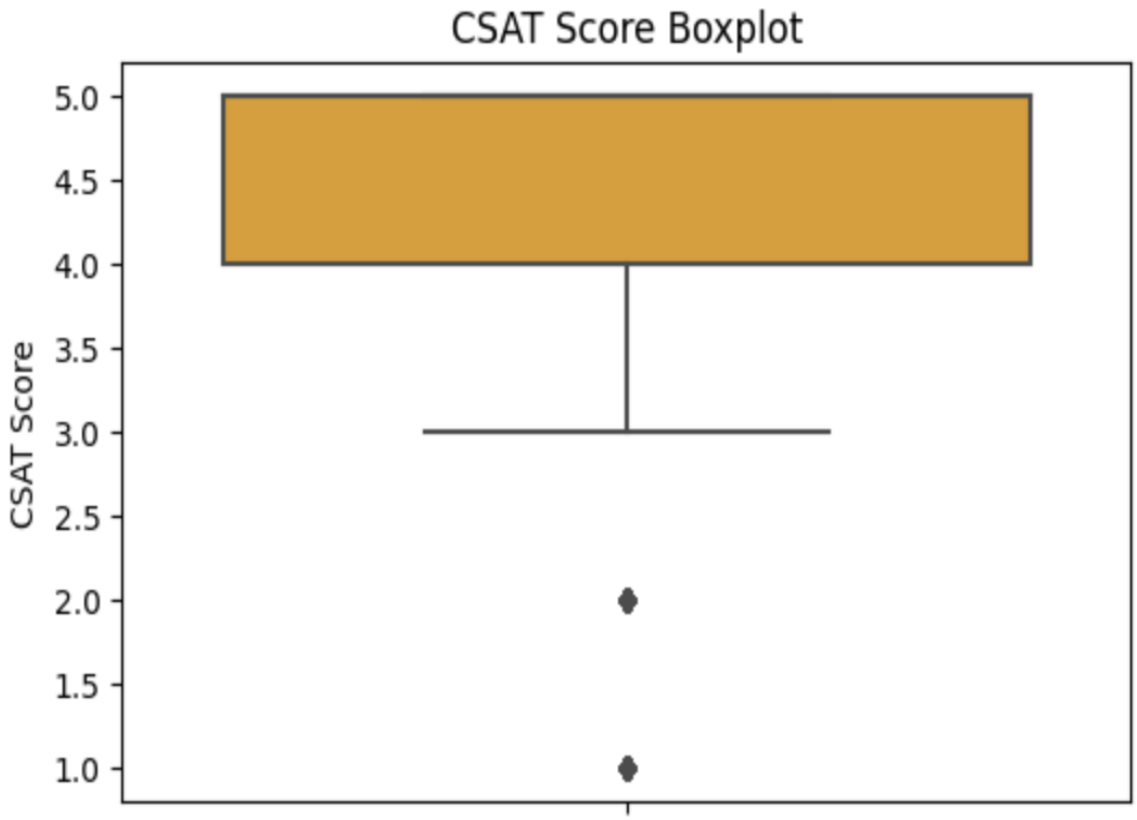

Fig.9.3 c : Boxplot of CSAT Score Boxplot

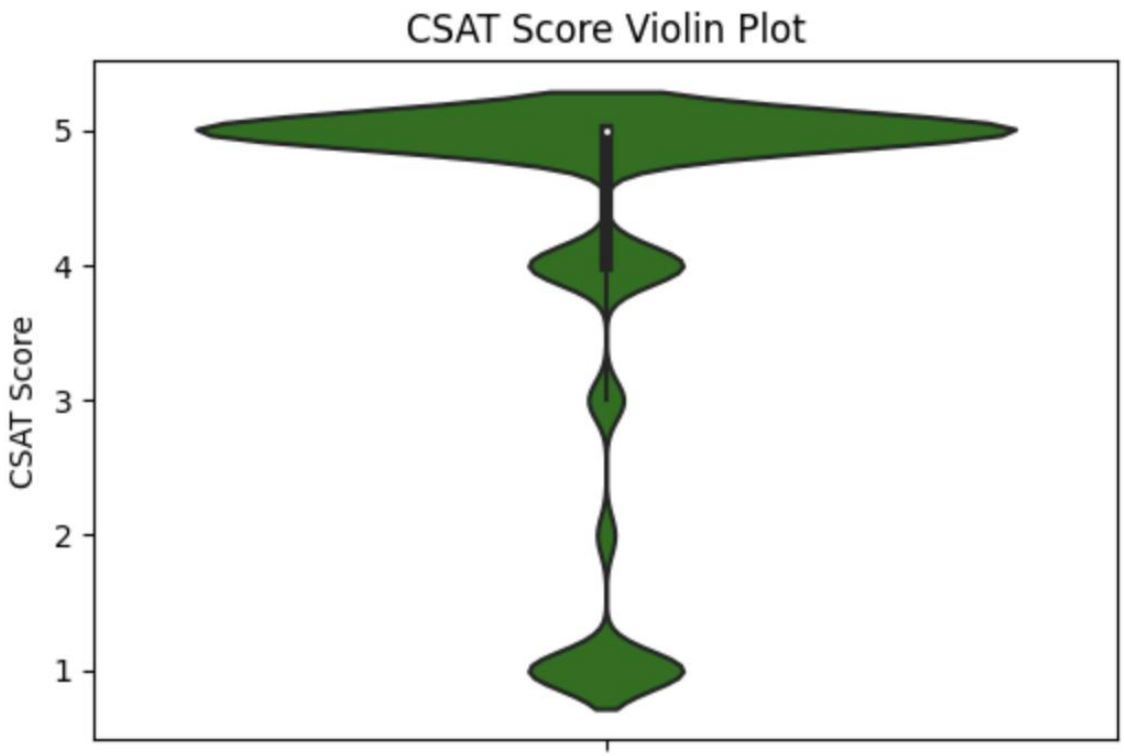

Fig.9.3 d: Violin plot of CSAT Score.

The Figure 4 violin plot of CSAT Scores provides a richer view of the distribution compared to a standard boxplot. It displays the probability density of the scores at different values, with the width of the "violin" indicating the density. This visualization clearly shows that CSAT scores are predominantly clustered at higher values (likely 5), with smaller densities observed at lower satisfaction levels, indicating a generally positive customer sentiment but also revealing the presence of dissatisfied customers.

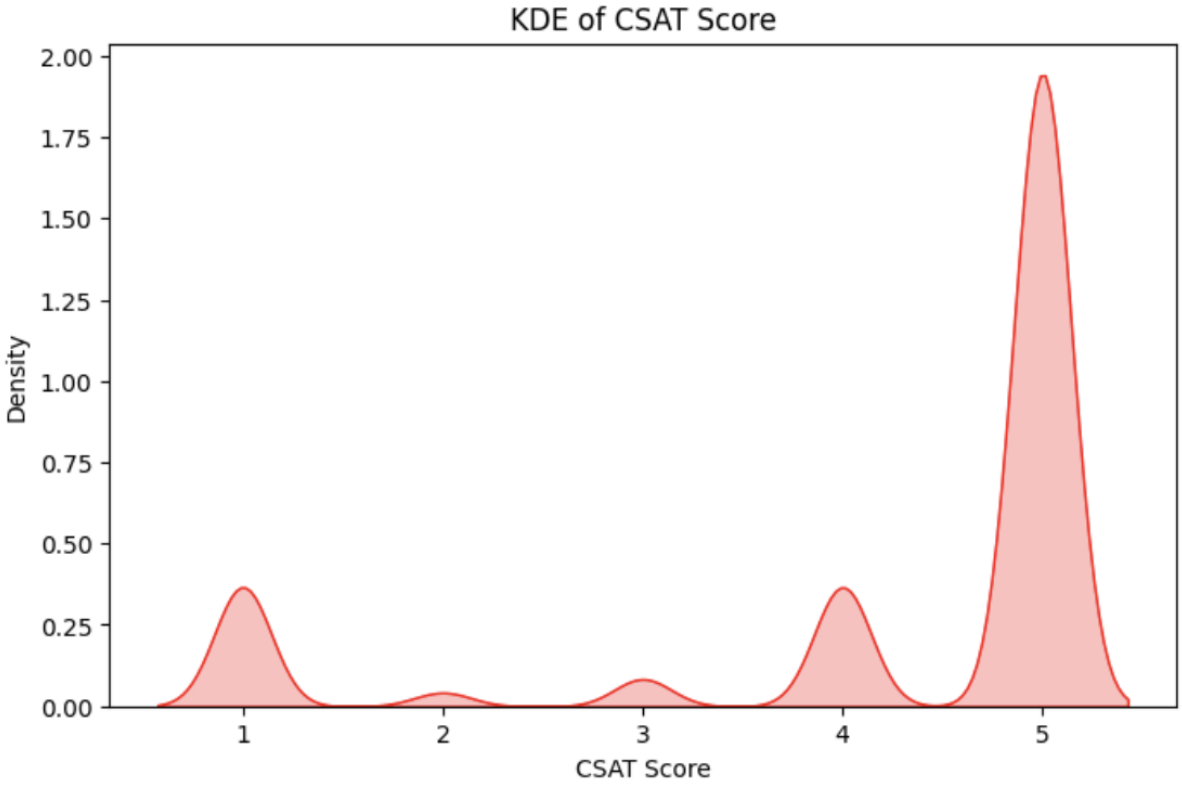

Fig.9.3 e: KDE plot of CSAT Score.

The KDE (Kernel Density Estimation) plot of CSAT Scores provides a smoothed estimate of the probability density function for customer satisfaction ratings. This visualization clearly shows multiple distinct peaks, indicating a bimodal or multimodal distribution. The most prominent peak is at a CSAT score of 5, signifying a high proportion of highly satisfied customers, while smaller peaks suggest other common satisfaction levels, including potentially dissatisfied groups.

The Figure 6 bar plot titled "CSAT Score Count" visually presents the frequency of each individual CSAT score received. It clearly shows that a CSAT score of '5' is by far the most frequent rating, followed by '1'. Scores of '2', '3', and '4' have significantly lower counts, indicating a highly polarized customer sentiment with a large majority of highly satisfied customers and a substantial segment of highly dissatisfied ones.

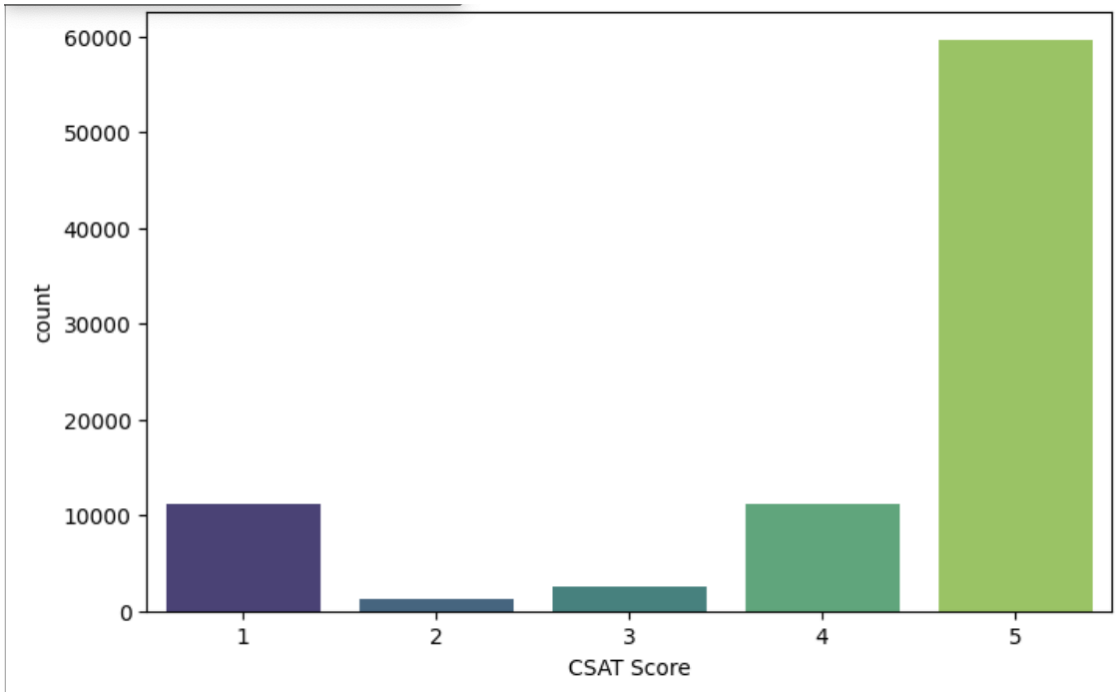

Fig.9.3 f: Bar plot of CSAT Score Count.

Figure 7 time series plot titled "CSAT Score Over Time" illustrates the average Customer Satisfaction (CSAT) score's fluctuations across different dates. The X-axis likely represents time (dates), while the Y-axis shows the average CSAT score. This visualization helps in identifying trends, seasonality, or specific periods where customer satisfaction significantly increased or decreased, allowing for a temporal analysis of service performance.

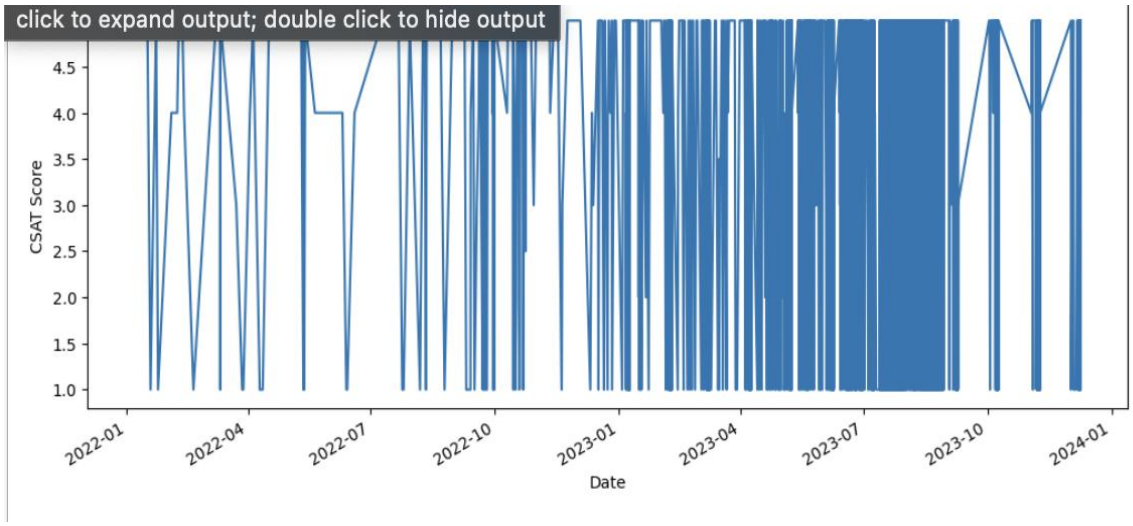

Fig.9.3 g: Time series analysis CSAT Score Over Time

.

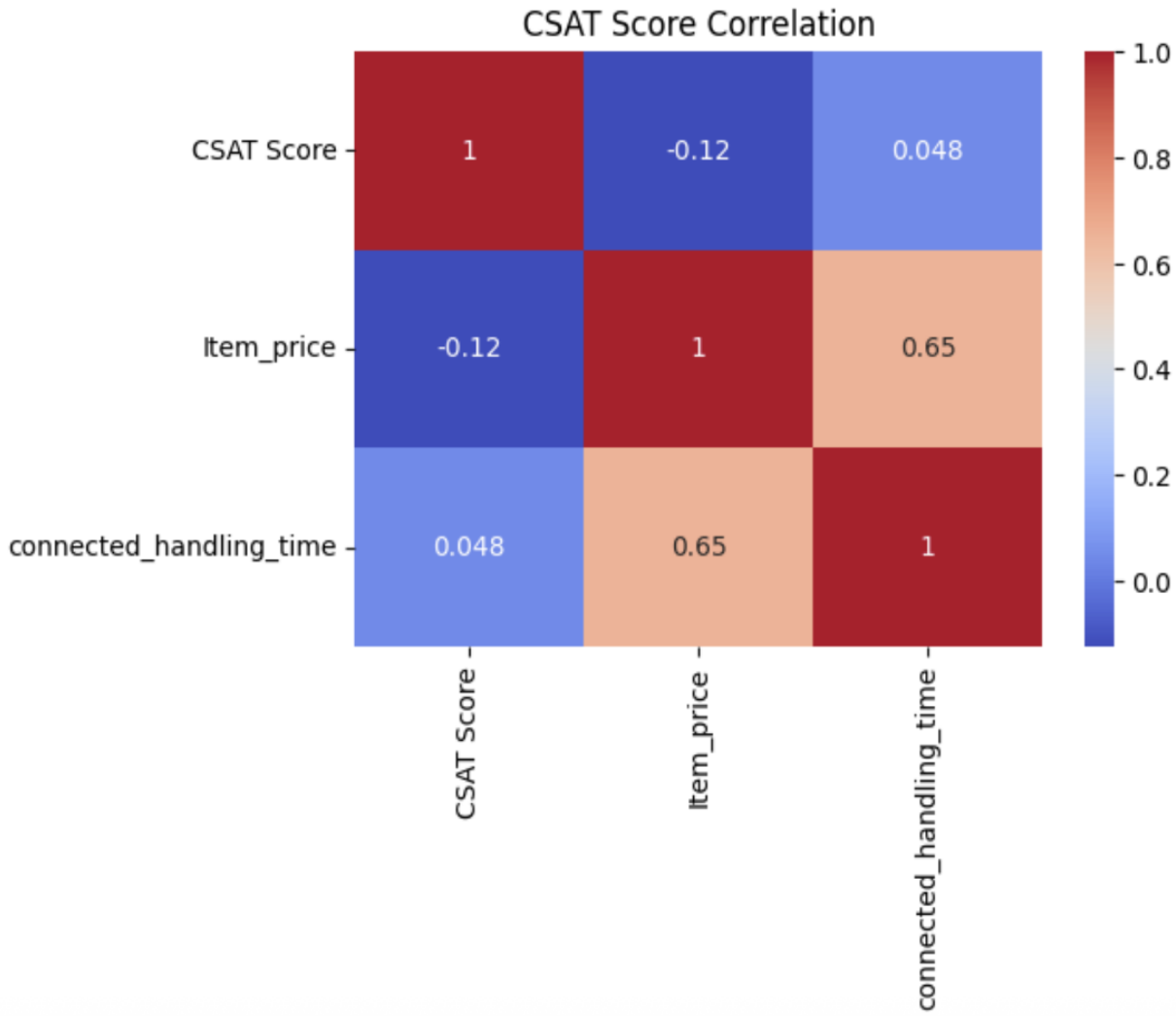

Fig 9.3 h: Heatmap of CSAT Score by Category

The heatmap titled "CSAT Score by Category" visually represents the distribution of CSAT scores across various categories. Each cell in the heatmap corresponds to a specific combination of `category` and `CSAT Score`, with color intensity indicating the concentration or frequency of responses. This visualization allows for quick identification of which categories are associated with higher or lower CSAT scores, highlighting areas of strong satisfaction or dissatisfaction within specific product or service categories.

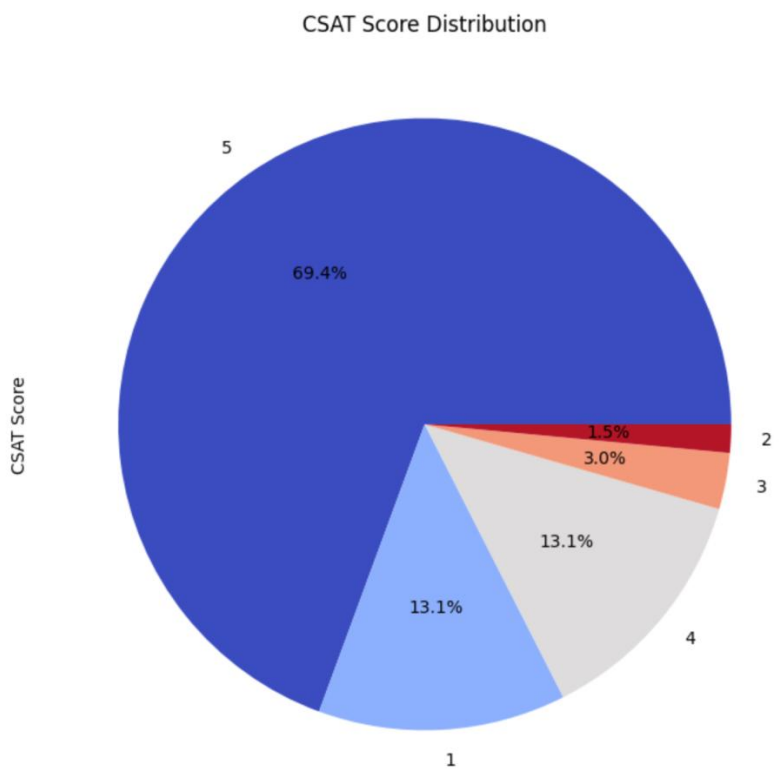

Fig.9.3 I:Pie Chart of CSAT Score Distribution.

The pie chart titled "CSAT Score Distribution" visually represents the percentage share of each individual CSAT score. It offers an immediate and intuitive understanding of the overall customer sentiment. The chart typically highlights that the largest proportion of customers give a '5' (highly satisfied) score, while '1' (highly dissatisfied) forms the second largest segment, underscoring a polarized distribution of customer satisfaction.

Figure 10 Boxen plot, also known as a letter-value plot, for "CSAT score by agent name" offers a detailed view of the CSAT score distribution for each individual agent. Unlike standard boxplots, the boxen plot provides more quantiles, particularly in the tails of the distribution, making it better for large datasets. This visualization helps in assessing not just the median performance of each agent, but also their consistency and the presence of extreme satisfaction or dissatisfaction scores linked to their interactions, aiding in performance evaluation and targeted training.

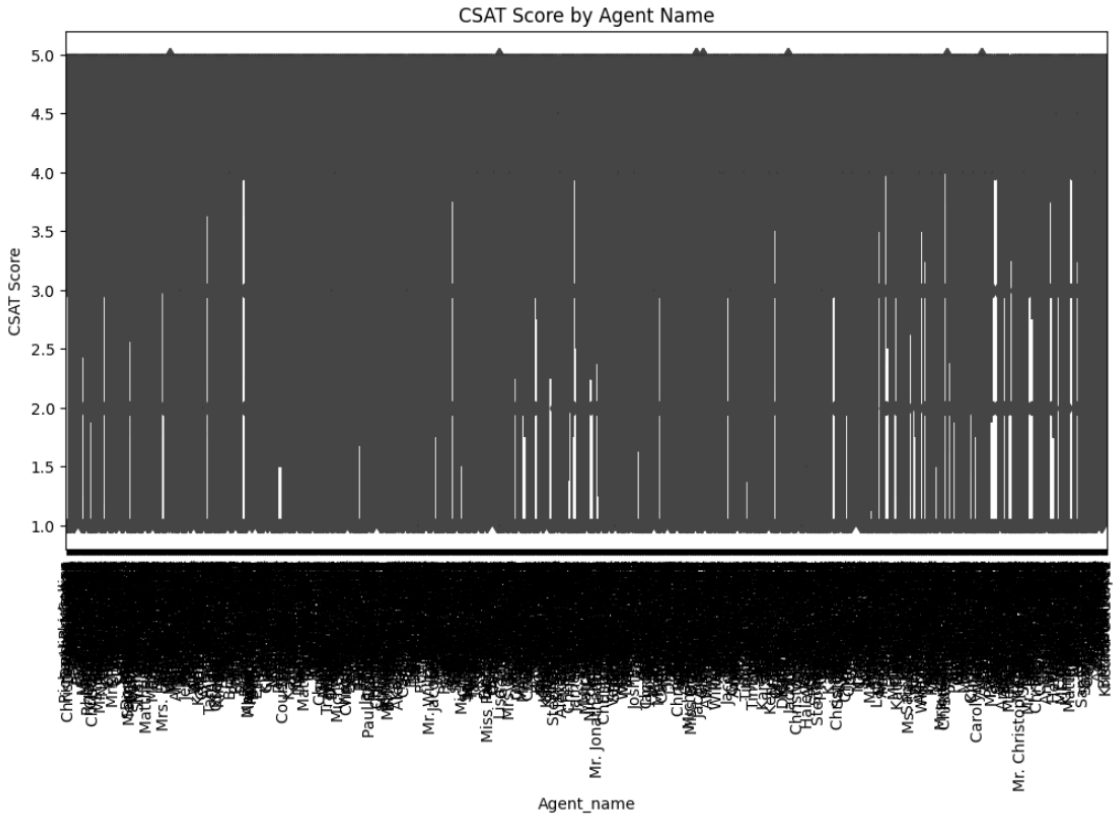

Fig. 9.3 j: Boxen plot of CSAT score by agent name.

Figure 11 ECDF (Empirical Cumulative Distribution Function) plot of CSAT scores displays the proportion of data points that fall below or at each given CSAT score value. It is a non-parametric way to visualize the distribution, showing the cumulative probability. From this plot, one can observe how the percentage of customers increases as the CSAT score increases, providing insight into the overall satisfaction level and the cumulative percentage of responses at each score point.

Figure 12 ridgeline plot titled "CSAT Score by Category" displays overlapping density plots for CSAT scores, separated by product category. This visualization allows for a clear comparison of the distribution of customer satisfaction across different item types. By showing the shape, peaks, and spread of scores for each category on a common axis, it helps identify which categories tend to have higher or lower satisfaction, or if satisfaction is more consistently distributed within certain categories.

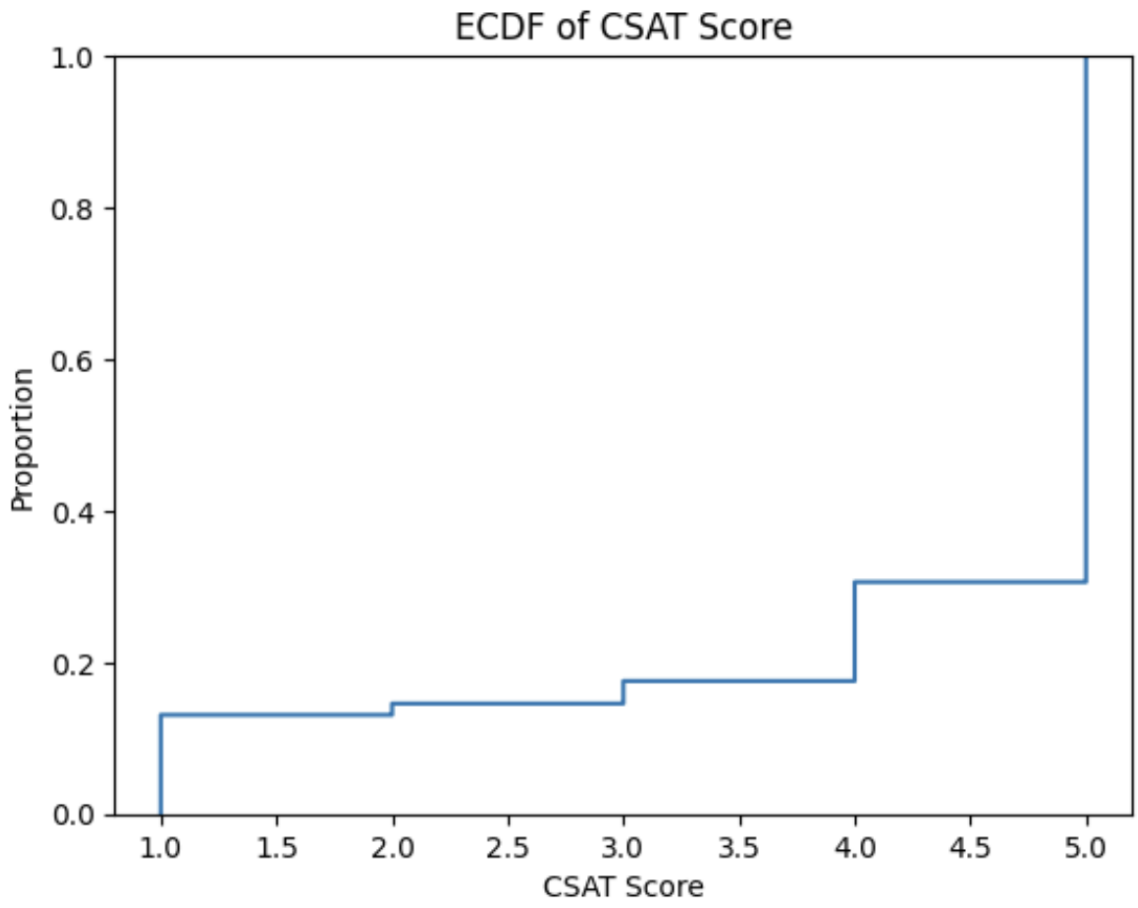

Fig. 9.3 k: ECDF plot of CSAT score.

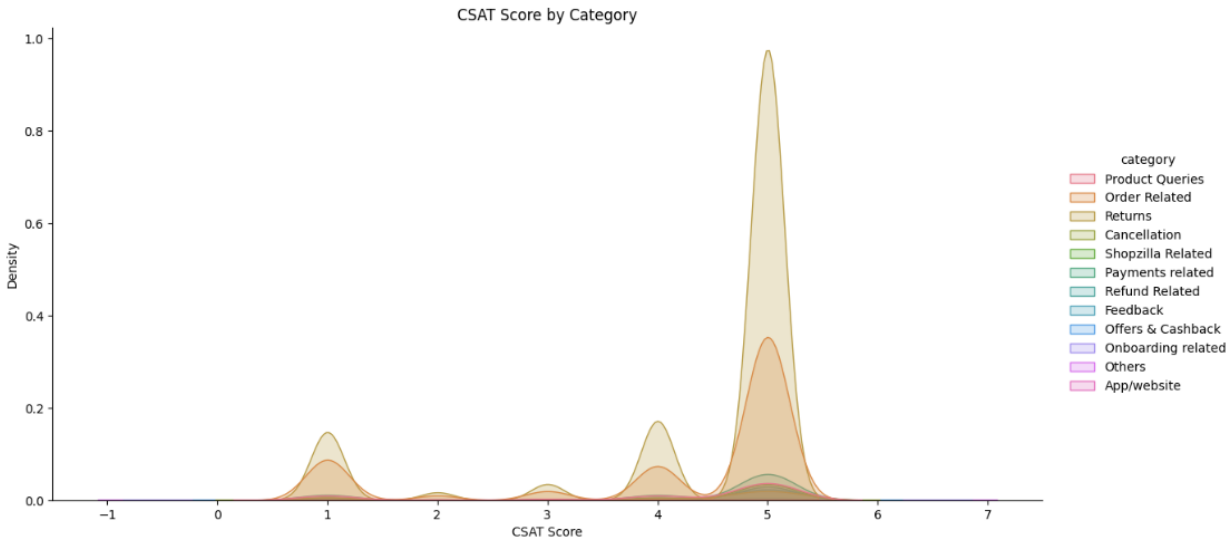

Fig.9.3 l: Ridgeline plot CSAT Score by Category.

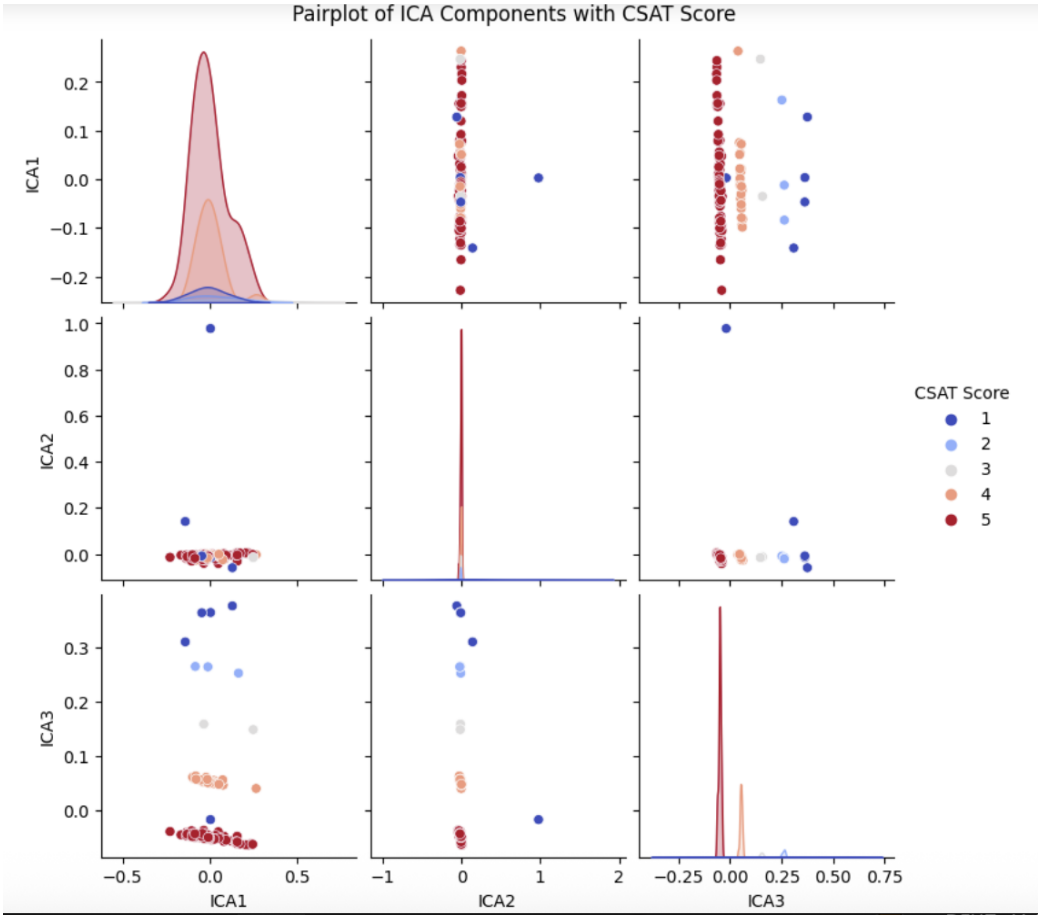

Fig.9.3 m: Pair plot of ICA Components with CSAT Score.

Figure 13 pair plot displays scatterplots for every pair of Independent Components (ICA1, ICA2, ICA3) derived from FastICA, along with the distribution of each component on the diagonal. Critically, these plots are colored by the CSAT Score, allowing for visual inspection of how different satisfaction levels cluster or spread across the independent feature space. This visualization is key for understanding non-linear relationships and identifying latent factors that distinctly influence customer satisfaction, beyond simple correlations.

CHAPTER 10
CONCLUSION AND FUTURE SCOPE

10.1 Conclusion

This project successfully explores the customer support dataset through comprehensive visualizations and statistical techniques, with a primary focus on understanding patterns and factors influencing the CSAT (Customer Satisfaction) Score. By utilizing plots like histograms, boxplots, KDEs, and correlation heatmaps, we identified key relationships between customer satisfaction and variables such as item price, handling time, and agent performance. Additionally, FastICA was applied to extract underlying independent components, offering deeper insights into data structure and aiding in dimensionality reduction. This thorough EDA lays a strong foundation for advanced analytics and decision-making.

10.2 Future Scope

The analysis can be extended by applying machine learning models to predict CSAT scores based on the identified features. Real-time monitoring dashboards and customer churn prediction systems can also be developed to enhance proactive customer service strategies.

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